INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

CAPSTONE PROJECT REPORT

FINAL REPORT · FINAL VERSION

Multi-Domain Agentic RAG Assistant

A LangGraph multi-agent, hybrid-retrieval question-answering system over five RAGBench domains, with evidence-grounded answers and a Safety Agent audit

Biomedical | Legal | Finance | Customer Support | General Knowledge

GROUP 2 · PROFESSIONAL CERTIFICATION PROGRAM IN ARTIFICIAL INTELLIGENCE, IIIT HYDERABAD

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Abstract

This project builds a **multi-domain, agentic Retrieval-Augmented Generation (RAG) assistant** that answers questions from five professional domains — biomedical research, legal contracts, corporate finance, IBM technical support and general knowledge — using only evidence retrieved from a knowledge base and citing every statement. The knowledge base is built from the public **RAGBench** benchmark (200 rows per domain, 3,211 unique documents, 14,704 chunks). A **LangGraph** workflow of five agents — the same for all five domains (Router Agent, Retrieval Agent, Domain Check Agent, Answer Agent and Safety Agent) — runs on **Groq** with the open-weight model `openai/gpt-oss-20b`.

Retrieval combines dense vectors (`all-MiniLM-L6-v2`, chosen over MiniLM-L12 and bge-small in a 3-model study) and BM25 keyword search with **Reciprocal Rank Fusion**, searches the routed domain first with a two-chunk global back-stop, corrects mis-routed questions with a retrieval-based domain vote, searches inside a single contract when one is named, and adds domain-specific evidence (finance report tables, error-code matches, a second-hop search for multi-hop questions). Chunk size, overlap, embedding model, fusion, top-k and re-ranking were chosen in a controlled 95-question retrieval experiment.

The system was developed in ten measured versions (v2.0 → **v2.9**, final). On the same 30-question test, correctness rose from **15/30 (50%)** in v2.1 to **25/30 (83%)** in v2.9, with routing 30/30, retrieval hit 28/30 and 29/30 answered with citations. After removing two questions whose RAGBench reference answers are faulty the score is **89%**, and **96%** on the answerable questions. Finance, the weakest domain, improved from 1/6 to 3/6. A further 18 in-dataset questions scored 12/18 correct and 3 partial. An experiment with the 6× larger `gpt-oss-120b` (v3.0 test) gave the same 25/30 at twice the cost, showing that the remaining errors come from the data, not from the model size. The system runs end-to-end in Google Colab with a Gradio chat interface.

Keywords: retrieval-augmented generation, multi-agent systems, LangGraph, hybrid search, BM25, sentence embeddings, reciprocal rank fusion, RAGBench, LLM-as-judge, AI safety guardrails.

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1. Introduction

1.1 Problem statement

Large language models answer fluently but can invent facts, cannot cite their sources and know nothing about private or recent documents. In professional domains — medicine, contracts, finance, technical support — an unsupported answer is worse than no answer. Retrieval-Augmented Generation (RAG) addresses this by retrieving relevant passages first and asking the model to answer only from them. A single RAG index over many domains, however, introduces new problems: questions are routed to the wrong domain, near-identical documents (e.g. 161 contracts with the same clause headings) compete for the same query, and domain-specific safety rules (no medical dosing, no legal or investment advice) must be enforced.

1.2 Objectives

- Build one assistant that serves **five domains** from a single, citable knowledge base.
- Design a **hybrid retrieval** pipeline (dense + keyword) and select its parameters — chunk size, overlap, embedding model, fusion and top-k — by **measurement**, not by default.
- Orchestrate the answer as a **multi-agent workflow** (LangGraph) with explicit routing, retrieval, domain-check, answer and safety agents.
- Enforce **grounding and safety**: every statement cites a retrieved source, unknown citations are removed, unsupported claims are removed, emergencies are escalated.
- **Evaluate** routing, retrieval hit-rate, answer rate and correctness per domain on a fixed test set, and improve the weakest domain version by version.
- Run on **low-cost infrastructure** (Google Colab, Groq, open-source models).

1.3 Scope and deliverables

Deliverable	Description
Colab notebook	`MultiDomain_RAG_Agents_Colab.ipynb` (v2.9, final) — installs, indexes, builds the agents and launches the chat UI
Chat application	Gradio interface with a public share link; optional contract picker and medication list
Test results	30-question test run for every version (v2.1 → v2.9) plus 18 in-dataset questions (from v2.5); Test Report document
Parameter study	Retrieval experiment over chunking, embedding models and retrieval methods (Sections 4–6)
This report and the presentation	Design, parameters, version history, results and lessons learned

Table 1: *Deliverables.*

2. Dataset and Domain Representation

2.1 RAGBench

RAGBench (`galileo-ai/ragbench`, Hugging Face, CC BY 4.0) is a benchmark of 100k RAG examples drawn from industry corpora. Each row holds a question, the source documents a retriever returned for it, and a reference answer. This project uses five of its sub-datasets, one per domain. Only the **documents** field is indexed; the question and answer fields are never put into the knowledge base, so the assistant cannot "look up" the reference answer.

2.2 The five domains

Domain	RAGBench subset	Source documents	Train rows	Rows used	Unique docs	Avg doc length (chars)	Chunks	Share
Biomedical	covidqa	CORD-19 COVID-19 research-paper passages	1,252	200	713	599	713	4.8%
Legal	cuad	Full commercial contracts from SEC EDGAR (CUAD)	1,530	200	161	54,338	8,430	57.3%
Finance	finqa	Company annual reports (10-K) — text and tables	12,502	200	568	1,393	1,043	7.1%
Customer support	techqa	IBM technotes and security bulletins	1,192	200	973	3,601	3,701	25.2%
General	hotpotqa	Wikipedia paragraphs (multi-hop questions)	1,883	200	796	487	817	5.6%
Total	5 subsets			1,000	3,211		14,704	100%

Table 2: *Domains, sources and index size (200 rows per domain, 1,200-character chunks with 150 overlap).*

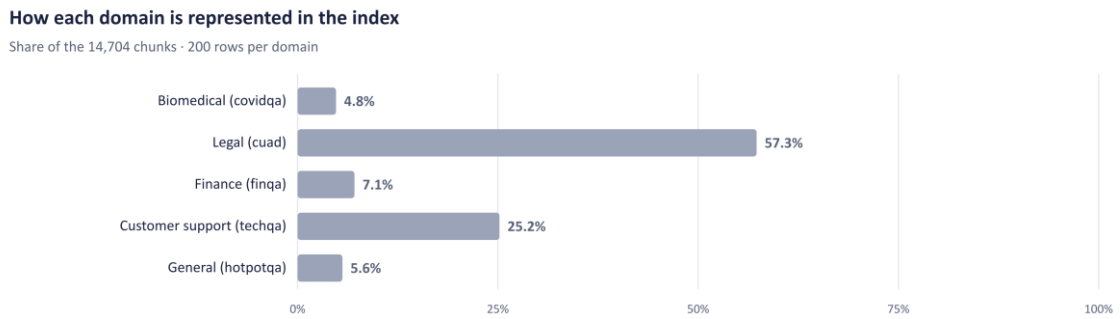


Figure 1: *Share of the index per domain — legal contracts dominate by volume.*

2.3 How each domain is represented

Domain	Unit of text	Chunks per document	What this means for retrieval	Domain rule given to the LLM
Biomedical	Short paper passage	1.0	A passage usually fits in one chunk — the full context is kept.	No diagnosis, prescription or patient-specific dose
Legal	Whole contract (~54k chars)	52.4	Each contract becomes ~52 chunks; 161 contracts share the same clause words, so a question must name the contract (Section 6.4); a named contract gets 12 chunks.	No legal advice; describe only what the contract says
Finance	Report text + table (separate document)	1.8	FinQA stores a report's table as a separate document; numbers for a calculation can sit in another chunk, so the report's table chunks are added (v2.5).	No investment/tax advice; show calculations step by step
Customer support	Technote page	3.8	Long pages split into ~4 chunks; product names and error codes make BM25 very effective (2 extra code matches, v2.5).	Only steps, commands and versions present in the excerpts
General	Wikipedia paragraph	1.0	Short paragraphs, one chunk each; multi-hop questions need two paragraphs (second-hop search, v2.8).	Answer only from the excerpts

Table 3: *Representation of each domain in the knowledge base.*

Every chunk carries metadata: a stable citation ID (`RAG-00001` …), its **domain** tag (used for domain filtering), its source (sub-dataset, row, document and chunk number), a `doc_key` identifying the parent document and, for contracts, a readable **contract title** such as *"Moelis — Strategic Alliance Agreement"*. Repeated passages are de-duplicated so the same text cannot fill several evidence slots.

2.4 Domain-wise results: first and final version

How a domain is represented decided how hard it was and which fix helped. The table links the representation to the measured results of the same six questions per domain in v2.1 and in the final v2.9.

Domain	Share of index	Chunks / doc	Hit v2.1	Hit v2.9	Correct v2.1	Correct v2.9	What made the difference
Biomedical	4.8%	1.0	5/6	6/6	3/6	4/6	Hybrid search finds the passage; the miss is one faulty reference (BIO-03) and one run-to-run variation (BIO-02).
Legal	57.3%	52.4	0/6	6/6	2/6	6/6	Contract-scoped search (v2.2) and 12 chunks for a named contract (v2.8).
Finance	7.1%	1.8	4/6	4/6	1/6	3/6	Report tables added (v2.5), table values labelled (v2.6), computed numbers accepted by the audit (v2.7). Remaining misses: no company named or faulty reference.
Customer support	25.2%	3.8	6/6	6/6	4/6	6/6	top-8 (v2.4) and error-code matches (v2.5) found the fix passages.
General	5.6%	1.0	6/6	6/6	5/6	6/6	Second-hop search for the missing fact (v2.8).
Total	100%	4.6	21/30	28/30	15/30	25/30	+10 correct answers

Table 4: *Domain representation and domain-wise results, v2.1 vs v2.9 (30-question test).*

3. System Architecture

The system has two halves: an **offline** pipeline that builds the knowledge base once per Colab session, and an **online** LangGraph workflow that answers each question through five agents (Figure 2). The same five agents serve all five domains.



Figure 2: *End-to-end architecture: offline indexing (left) and the five-agent online workflow (right).*

3.1 Technology stack

Layer	Technology	Role
Orchestration	LangGraph 1.x (StateGraph)	Five-node agent workflow with typed shared state
LLM	Groq API · `openai/gpt-oss-20b` (temperature 0)	Router, Answer and Safety Agents
LLM judge (evaluation only)	Groq · `openai/gpt-oss-120b`	Grades answers against the RAGBench reference
Evaluation	30 test questions (6 per domain), results in Excel	Answers checked against RAGBench reference answers
Embeddings	sentence-transformers `all-MiniLM-L6-v2` via langchain-huggingface	384-dimensional dense vectors
Vector store	LangChain InMemoryVectorStore (Colab) / Chroma (persistent)	Cosine similarity search with domain filter
Keyword search	rank-bm25 (BM25Okapi)	One index per domain + one global
Fusion	Reciprocal Rank Fusion, score 1 / (60 + rank)	Merges dense and BM25 results; top 8 chunks (12 for a named contract)
Data	Hugging Face `datasets` · RAGBench	Five domain corpora
Chunking	Own splitter: 1,200 characters with 150 overlap	14,704 chunks across the five domains
External API	OpenFDA drug-label API	Domain Check Agent (health only)

Layer	Technology	Role
Validation	Pydantic models + JSON-schema structured output	Typed agent outputs, retries on malformed JSON
Interface	Gradio ChatInterface (share link)	Chat, contract picker, medication list
Web app	Python ThreadingHTTPServer (web_app.py) + HTML / JavaScript chat page	Login-protected demo app; API key kept in .env
Runtime	Google Colab (T4 GPU recommended)	Hosting of the whole pipeline
Code storage	GitHub (private repositories)	Version control for the notebook and the web app

Table 5: *Technology stack.*

3.2 Request flow

1. The user asks a question in the Gradio chat (optionally picks a contract or lists medications).
2. **Router Agent** (`router`) classifies the domain, severity and emergency risk, and extracts any contract name or medication names.
3. **Retrieval Agent** (`retrieval`) runs hybrid BM25 + vector search and selects up to eight evidence chunks (contract-scoped, domain-first, with domain vote and back-stop) and adds domain-specific evidence.
4. **Domain Check Agent** (`domain_check`) runs the domain's extra check: for health questions with medications it fetches OpenFDA labels and keeps only verbatim-verified findings; the other domains need nothing extra.
5. **Answer Agent** (`answer`) writes the answer from the excerpts only, citing each statement with `[RAG-xxxxx]` IDs.
6. **Safety Agent** (`safety`) verifies every statement: it removes unknown citations, removes claims that two audits find unsupported, applies emergency rules and sets the expert-review flag.
7. The UI shows the answer, the cited sources, a **Router Agent:** line (domain, severity, contract), a **Domain Check Agent:** line and any safety notes.

4. Data Ingestion and Chunking

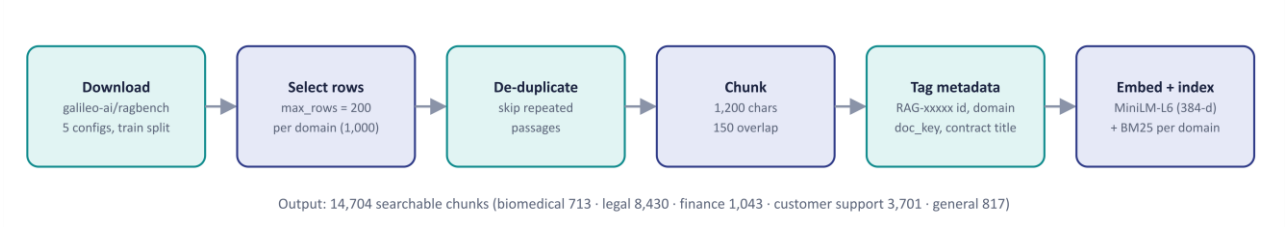


Figure 3: *Ingestion and indexing pipeline.*

4.1 Chunking algorithm

Documents are split into windows of at most `chunk_size` characters. If a window would end mid-text, the cut moves back to the **last space in the second half of the window**, so words are never broken. The next window starts `overlap` characters before the previous end, so a sentence that straddles a boundary appears in both chunks. Character-based chunking was chosen over token-based chunking because it is model-independent, deterministic and fast; at roughly 4 characters per token, 1,200 characters ≈ 250–300 tokens, which fits the 256-token input window of MiniLM with little truncation.

4.2 Chunking experiment

Seven size/overlap settings were compared on the same corpus and the same 95 questions (20 per domain, 15 legal), using the production hybrid search and MiniLM embeddings. A question counts as a **hit** when at least one of the five retrieved chunks comes from that question's own source documents; MRR is the mean reciprocal rank of the first such chunk. The experiment needs no LLM calls, so it is free. Experiment corpus: covidqa, finqa and hotpotqa 100 rows each, techqa 50 rows, cuad 15 contracts.

Chunk size	Overlap	Overlap %	Chunks	Avg chars	Embed time (CPU)	Bio	Legal	Finance	Support	General	Hit@5	MRR
300	50	17%	11,620	278	385 s	90%	20%	90%	100%	100%	83%	0.77
300	150	50%	18,481	286	781 s	95%	13%	95%	100%	100%	84%	0.80
500	100	20%	7,300	455	418 s	95%	13%	100%	100%	100%	85%	0.82
800	150	19%	4,645	695	301 s	90%	20%	95%	100%	100%	84%	0.79
1200	150	13%	3,236	933	163 s	95%	20%	90%	100%	100%	84%	0.80
1200	300	25%	3,535	960	241 s	95%	13%	90%	100%	100%	83%	0.81
2000	200	10%	2,332	1259	158 s	95%	20%	95%	100%	100%	85%	0.79

Table 6: *All seven chunk size / overlap settings with hit@5 per domain (highlighted row = chosen).*

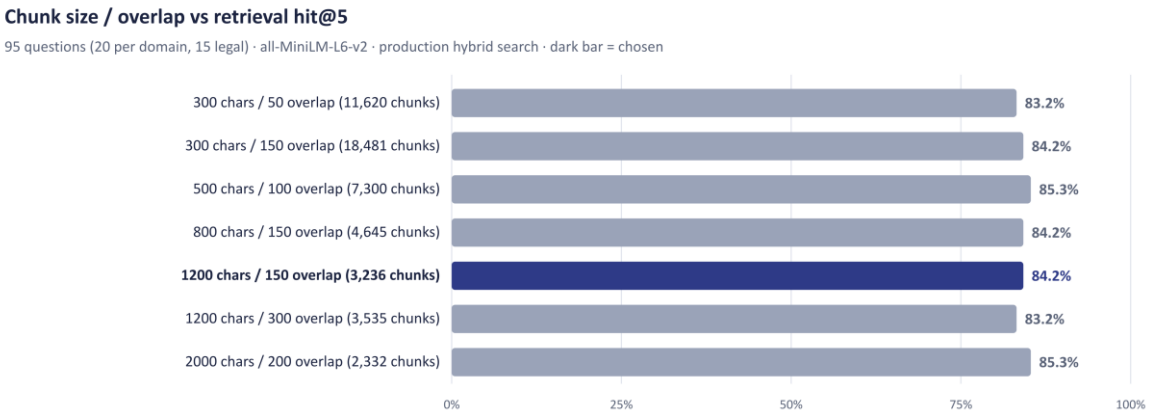


Figure 4: *Retrieval hit@5 by chunking setting.*

4.3 300 / 150 versus 1,200 / 150

The smallest setting with a large overlap (300 characters, 150 overlap = 50%) was the main alternative. The table compares it with the chosen setting domain by domain.

Measure	300 / 150	1,200 / 150 (chosen)	Difference (300 vs 1,200)
Chunks (experiment corpus)	18,481	3,236	5.7× more
Average chunk length	286 chars	933 chars	
Embedding time (CPU)	781 s	163 s	4.8× longer
Context in 8 evidence chunks	≈ 2,288 chars	≈ 7,464 chars	much less evidence per answer
Hit@5 — Biomedical	95%	95%	same
Hit@5 — Legal	13%	20%	-7 pts
Hit@5 — Finance	95%	90%	+5 pts
Hit@5 — Customer support	100%	100%	same
Hit@5 — General	100%	100%	same
Hit@5 — overall	84%	84%	same
MRR	0.80	0.80	0.00
Full index (estimated)	≈ 83,975 chunks	14,704 chunks	

Table 7: *300 / 150 compared with 1,200 / 150, per domain.*

4.4 Why 1,200 characters with 150 overlap

- **Accuracy:** 1,200 / 150 reached **84% hit@5** (MRR 0.80); the best setting tried reached 85%. With 95 questions one question is ≈ 1 point, so every setting lies within two questions of the others — the choice is decided by cost, by the embedding model's input window and by how much context each answer receives.
- **Cost:** it has the fewest chunks of any setting that fits the embedding window (3,236 here, 14,704 for the full index) and among the shortest embedding times (163 s on CPU here) — smaller chunks multiply both.
- **Context per answer:** in the study (top-5), five 1,200-character chunks gave the LLM ≈ 4,665 characters of evidence — enough to hold a whole passage, a table or a clause; the final system's top-8 gives ≈ 7,464 characters, still well inside the model's context window.

- **Fits the domains:** biomedical and general passages (≈ 500–600 characters) stay whole, which keeps each answer and its context in one citable chunk; technotes split into ≈ 4 pieces and contracts into ≈ 52.
- **Fits the model:** ≈ 250–300 tokens per chunk is close to MiniLM's 256-token input window, so little text is truncated; 2,000-character chunks lose their tail to truncation.
- **Overlap 150 (12.5%):** enough to keep a sentence that crosses a boundary in both chunks; 300 overlap added chunks and embedding time without a measurable gain.
- **Smaller chunks (500 / 100):** 85% hit@5 (MRR 0.82) — one question better, but 2.3× the chunks, 2.6× the embedding time and only ≈ 2,275 characters of context per answer; legal fell to 13%.
- **Larger chunks (2,000 / 200):** 85% hit@5 with fewer chunks (2,332), but a 2,000-character chunk is ≈ 450 tokens while MiniLM reads only the first 256 — the second half of every chunk is invisible to dense search (BM25 still sees it, which is why the hit-rate holds). Larger chunks also make the hit metric easier and send up to ≈ 10,000 characters to each of the answer, audit and repair calls, which does not fit the free tier's 8,000 tokens per minute.

Why not 300-character chunks with 150 overlap?

300 / 150 reached 84% hit@5 — the same as 1,200 / 150 (84%) — but needed **5.7× the chunks** (18,481 vs 3,236) and **4.8× the embedding time** (781 s vs 163 s).

Scaled to the full index this is ≈ 83,975 chunks: several hours of embedding on Colab's CPU and a much larger in-memory index.

A 50% overlap means every sentence is stored about twice, so neighbouring near-duplicate chunks fill the evidence slots, and in the study (top-5) the LLM received only ≈ 1,430 characters of context instead of ≈ 4,665 — often too little for a calculation or a full clause. Legal retrieval also dropped (13% vs 20%).

5. Embedding Model Selection

5.1 Models tried and their results

Three sentence-embedding models were compared with everything else fixed (1,200 / 150 chunks, the same 3,236 chunks and 95 questions). Each was measured as dense-only search and inside the production hybrid search.

Model	Parameters	Dimensions	Max input tokens	Query / passage prefix	Embed time (CPU)	Dense hit@5	Hybrid hit@5	Hybrid legal	Hybrid MRR
all-MiniLM-L6-v2	22.7 M	384	256	none	163 s (1.0×)	84%	84%	20%	0.80
all-MiniLM-L12-v2	33.4 M	384	128	none	223 s (1.4×)	79%	82%	13%	0.76
BAAI/bge-small-en-v1.5	33.4 M	384	512	query instruction	818 s (5.0×)	85%	87%	33%	0.83

Table 8: *Embedding models compared on 3,236 chunks (1,200 / 150) and the same 95 questions; CPU timings on a 4-core laptop (highlighted = chosen).*

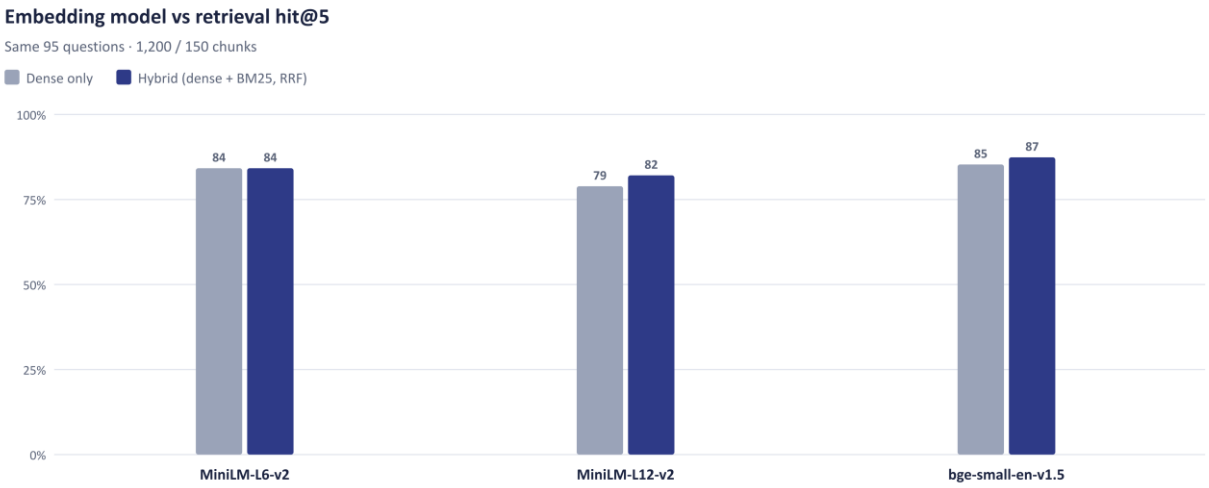


Figure 5: *Dense-only and hybrid hit@5 per embedding model.*

5.2 Embedding parameters used

Parameter	Value	Reason
model_name	`sentence-transformers/all-MiniLM-L6-v2`	See 5.3
normalize_embeddings	True	Unit vectors make dot product = cosine similarity
batch_size	256 on GPU, 32 on CPU (v2.3)	Fills a T4 GPU; avoids memory pressure on CPU
device	cuda if available, else cpu	Embedding ~14.7k chunks: ~2–3 min on T4 GPU vs ~1–1.5 h on Colab CPU
vector store	InMemoryVectorStore (Colab) / Chroma (local, persistent)	No server needed; Chroma caches the index between runs

Table 9: *Embedding configuration.*

5.3 Why all-MiniLM-L6-v2

- **Accuracy:** in the production hybrid search MiniLM-L6 scored **84%** and the best model 87% — a difference of 3 of 95 questions. Hybrid scores span 5 points across models versus 6 points for dense-only search (all-MiniLM-L6-v2 84%, all-MiniLM-L12-v2 79%, bge-small-en-v1.5 85%), because BM25 recovers many questions the dense model misses.
- **Fastest:** 163 s for 3,236 chunks on CPU; all-MiniLM-L12-v2 1.4×, bge-small-en-v1.5 5.0× slower. This matters on free Colab, where the index is rebuilt every session.
- **Smallest:** 22.7 M parameters and 384-dimensional vectors — the smallest memory footprint for the in-memory index of 14,704 chunks.
- **Proven default:** trained on 1 B+ sentence pairs for semantic search, symmetric (no query/passage prefixes), widely used with LangChain — easy to reproduce.
- **all-MiniLM-L12-v2** (twice the layers of L6, but its default input window is only 128 tokens, so it reads less than half of each 1,200-character chunk): hybrid 82%, dense 79%, 223 s — no gain over MiniLM-L6 for the extra cost.
- **bge-small-en-v1.5** (a newer retrieval-tuned model with a 512-token window; needs a query instruction prefix): hybrid 87%, dense 85%, 818 s — a gain of 3 points that does not justify 5.0× the embedding time on the free tier.
- **Recommendation:** bge-small-en-v1.5 is the most accurate model measured; switch to it when a GPU runtime is guaranteed (≈ 13 minutes for the full index on a T4, estimated from the measured ratio), keeping all other parameters unchanged.

5.4 The embedding model across versions

The embedding model was **kept unchanged from v2.0 to v2.9**. Every later accuracy gain came from what is retrieved around the embedding (top-k, contract scope, report tables, error-code matches, second-hop search) and from the Answer and Safety Agents — not from a different embedder. The study shows why: with BM25 fused in, the three models are within 5 points of each other, while the version changes in Chapter 9 moved the live score by 10 questions.

6. Retrieval System

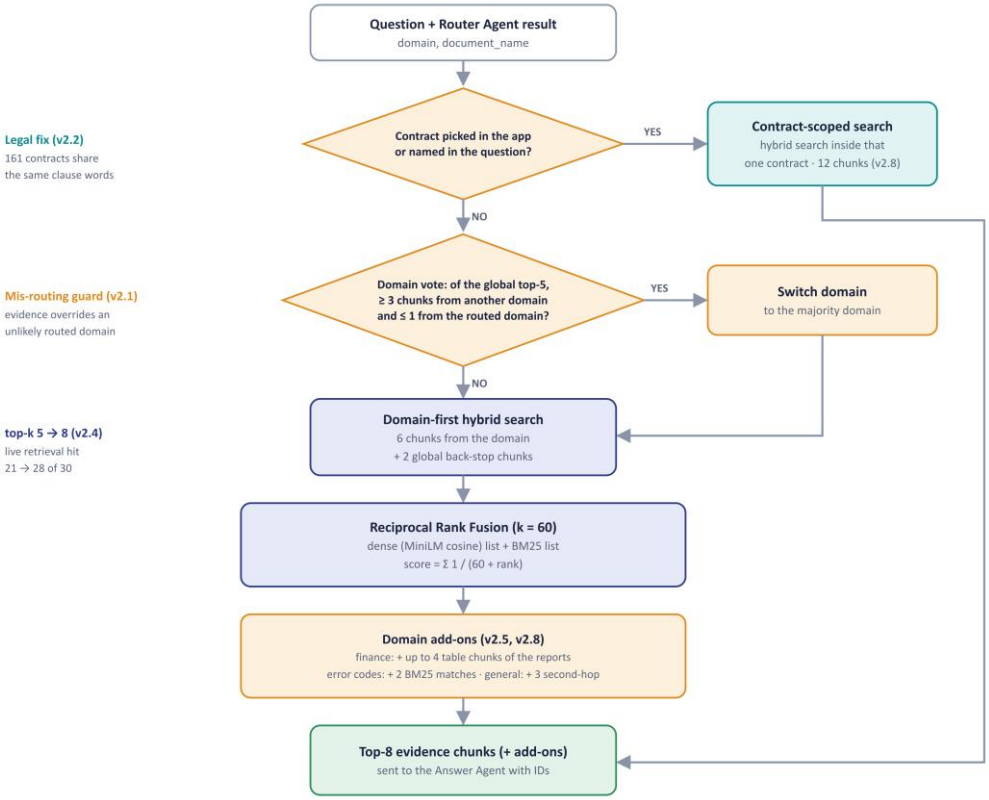


Figure 6: *Retrieval decision flow (v2.9).*

6.1 Hybrid search with Reciprocal Rank Fusion

Dense vectors capture meaning ("warranty duration" ≈ "guarantee period"), while BM25 rewards exact terms such as product names, error codes, drug names and years. Each searcher returns its own ranked list; **Reciprocal Rank Fusion** merges them without having to calibrate two incompatible score scales:

$$RRF(d) = \Sigma \text{ over lists } 1 / (k + rank(d)), \quad k = 60$$

A chunk ranked high by both searchers wins; a chunk ranked first by only one still scores well. k = 60 is the constant from the original RRF paper (Cormack et al., 2009) and dampens the influence of any single list.

6.2 Domain-first search with a global back-stop

Searching only the routed domain removes noise from the other four corpora, but a wrong routing decision would then return nothing useful. The Retrieval Agent therefore takes the **best 6 chunks from the routed domain plus the best 2 from all domains** (`TOP_K = 8`, `GLOBAL_BACKSTOP = 2`), and fills any shortfall from the domain list. Until v2.3 the split was 3 + 2 (top-5); the move to top-8 in v2.4 raised the live retrieval hit from 21 to 28 of 30 questions.

6.3 Retrieval-based domain vote

Before the domain search, a global search is run. If **at least 3 of its top 5 chunks** come from one other domain and **at most 1** from the routed domain (`DOMAIN_VOTE_MIN = 3`), the domain is switched and a note is shown. This corrected the routing mistakes seen in v2.0 (routing 81% → 100%), and routing has been 30/30 in every version since v2.1.

6.4 Contract-scoped legal search (v2.2, v2.8)

CUAD questions are generic ("Is there a non-compete clause?") and the 161 contracts share the same vocabulary, so a whole-domain search cannot know *which* contract is meant — legal hit@5 was 0/6 in the v2.1 test. The fix: the Router Agent extracts any contract or company name into `document_name`; `match_contract()` matches it against every contract's title and opening text (token overlap ≥ 0.6 and a unique best match); retrieval then runs the same hybrid RRF search **inside that one contract**. Users can also pick the contract from a drop-down. In the experiment, contract-scoped search reached **100%** legal hit@5 versus **20%** for the domain search. Because a contract's clauses are spread over about 50 chunks, v2.8 gives a named contract **12 chunks** (`LEGAL_TOP_K`) instead of 8.

6.5 Domain-specific evidence (v2.5, v2.8)

Add-on	Domain	Version	What it does	Measured effect
`REPORT_TABLES = 4`	Finance	v2.5	FinQA stores a report's table as a separate document; up to 4 table chunks of each retrieved report are added	Answer table now present for FIN-03, FIN-04; finance 1/6 → 2/6
Labelled table values	Finance	v2.6	Each table value is shown with its column label (e.g. "2016: 1,234")	Fixed the "after 2020" vs "total" mix-up (FIN-04)
`CODE_EXTRA = 2`	Support (any)	v2.5	Error codes and identifiers in the question (SQL1585N, CVE-…) get 2 extra BM25 matches, appended to the list	Found CUS-02's fix; fusing them into RRF instead pushed out good chunks
`FOLLOW_UP_K = 3`	General	v2.8	A second-hop search for the missing fact of a multi-hop question	GEN-02 finds "Gibraltar head of state"; general 5/6 → 6/6
`LEGAL_TOP_K = 12`	Legal	v2.8	More chunks from a named contract	Answer sentences found: IN-LEG-03 2/8 → 5/8, LEG-01 0/3 → 3/3

Table 10: *Domain-specific retrieval add-ons.*

6.6 Retrieval experiments

Retrieval method (top-5)	Bio	Legal	Finance	Support	General	Hit@5	MRR
BM25 only (domain)	95%	20%	95%	100%	100%	85%	0.80
Dense only (domain)	90%	20%	100%	95%	100%	84%	0.74
Hybrid RRF, no domain filter	90%	20%	95%	100%	100%	84%	0.79
Hybrid RRF, domain only	95%	20%	95%	100%	100%	85%	0.80
Hybrid RRF, domain + backstop 2	95%	20%	90%	100%	100%	84%	0.80
Hybrid, domain + backstop 1	95%	20%	90%	100%	100%	84%	0.80
Hybrid, domain + backstop 3	95%	20%	90%	100%	100%	84%	0.80
RRF k=10 (domain)	95%	20%	95%	100%	100%	85%	0.80
RRF k=30 (domain)	95%	20%	95%	100%	100%	85%	0.80
RRF k=60 (domain)	95%	20%	95%	100%	100%	85%	0.80
RRF k=100 (domain)	95%	20%	95%	100%	100%	85%	0.80
Hybrid top-20 + cross-encoder rerank	95%	13%	100%	100%	100%	85%	0.81

Table 11: *Retrieval methods, back-stop sizes and RRF constants tried (MiniLM, 1,200 / 150).*

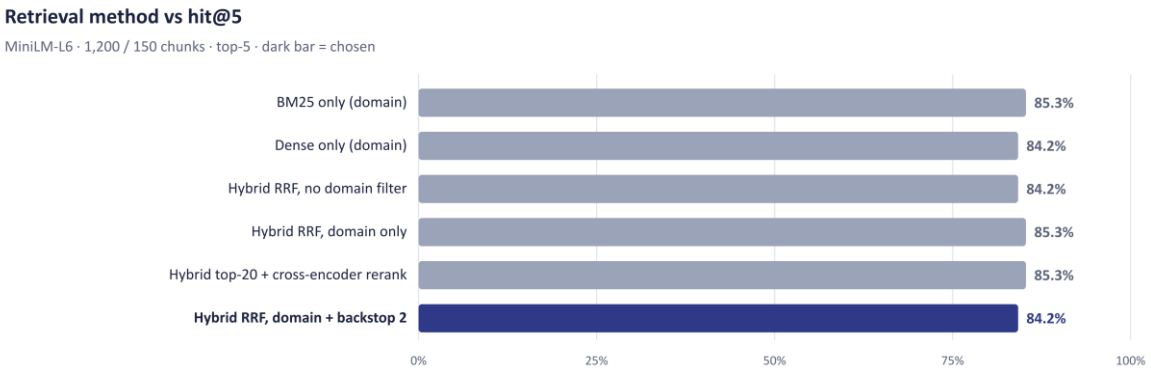


Figure 7: *Retrieval method comparison.*

top_k	Hit@k	MRR	Context sent to the LLM	Comment
3	83%	0.77	≈ 2,799 chars	Too little evidence; multi-part answers miss inputs
5	84%	0.80	≈ 4,665 chars	Used v2.0–v2.3: good recall within the free-tier token budget
8	87%	0.83	≈ 7,464 chars	Chosen from v2.4: higher recall; answer passage found for 28 of 30 live questions
10	90%	0.83	≈ 9,330 chars	Best recall, but double the tokens of top-5 for little gain over 8

Table 12: *Number of evidence chunks (top_k); 8 is used from v2.4.*

6.7 Why the other options were not chosen

- **BM25 only** (85%) and **dense only** (84%, MRR 0.74) fail on different questions — dense was better on finance (100% vs 95%), BM25 on biomedical (95% vs 90%) and support (100% vs 95%). Hybrid RRF stays within one question of the better searcher on every domain and gives the best ranking (MRR 0.80 vs 0.80 BM25 and 0.74 dense), so it is the most robust choice when the question type is unknown.
- **No domain filter** (84%, MRR 0.79) lets the other four corpora push relevant chunks down; filtering by domain improves ranking.
- **Domain only, no back-stop** scored 85% vs 84% with the 2-chunk back-stop in this test — but the test uses the true domain. In the live system the Router Agent can be wrong (v2.0 routed only 81% correctly); the back-stop guarantees two chunks from the right corpus even then. Back-stop 1 and 3 gave the same hit-rate, so 2 balances safety and in-domain depth.
- **RRF constant k** (10, 30, 60, 100) gave identical results on this set; k = 60 is kept as the standard value from the RRF paper.
- **Cross-encoder re-ranking** (ms-marco-MiniLM-L-6) reached 85% (MRR 0.81) but added 1.93 s per question on CPU and lowered legal retrieval (13%); not worth it on free hardware.
- **top_k:** 8 chunks raised hit@k to 87% and 10 to 90%, but every extra chunk is sent to the Answer Agent and auditor LLM calls. Top-5 was kept on the 8,000 tokens-per-minute free tier until v2.3; v2.4 moved to **top-8**, and the live retrieval hit rose from 21 to 28 of 30 questions. A named contract gets 12 chunks (v2.8), because its clauses are spread over about 50 chunks.
- **Contract-scoped search** is the one change with a large effect: legal hit@5 rose from 20% to **100%** when the contract is known.

7. Multi-Agent Pipeline (LangGraph)

The workflow is a LangGraph `StateGraph` over a typed state (`AssistantState`): each node reads the state, adds its output, and passes it on. The same five agents serve all five domains; the Domain Check Agent only does extra work for health questions (OpenFDA medication labels) and passes the other domains straight on. Structured agents return Pydantic models through JSON-schema output with up to three retries, because the model occasionally echoes the schema instead of filling it. Since v2.9 an empty model reply (gpt-oss occasionally spends its whole output budget on reasoning) is retried with `reasoning_effort = low`.

#	Agent (node)	LLM call	Input	Output	Key settings
1	**Router Agent** `router`	Yes (TriageResult)	Question, user medications	domain, document_name, category, severity, emergency, clinician_review, medications, rationale	6 domain labels; 4 severity levels
2	**Retrieval Agent** `retrieval`	No (general: 1 second-hop query)	Question, routing result, picked contract	Up to 8 evidence chunks (12 for a named contract) + domain add-ons, notes	top_k 8 · back-stop 2 · vote ≥ 3 · RRF 60 · contract match ≥ 0.6
3	**Domain Check Agent** `domain_check`	Only for health questions with medications	Domain, medication names	Health: OpenFDA labels, verified findings; other domains: nothing extra	≤ 6 medications · 3,000 label chars · quote must match label
4	**Answer Agent** `answer`	Yes (text)	Question, evidence, labels	Cited draft answer	Domain rule · answer first · one ID per bracket · empty reply → retry (low effort)
5	**Safety Agent** `safety`	Yes (2 × SafetyAssessment) + optional repair	Draft, evidence	Final answer, review flag, notes	Unknown IDs removed · two audits · repair · 6 emergency patterns

Table 13: *The five agents.*

7.1 Domain rules

Domain	Rule injected into the Answer Agent and audit prompts
Biomedical	Do not diagnose, prescribe, recommend starting/stopping/changing medicine, or give a patient-specific dose.
Legal	Do not give legal advice or say what anyone should sign or do; describe only what the contract excerpts say.
Finance	Do not give investment, tax or trading advice; show any calculation step by step using only numbers in the excerpts.
Customer support	Give only steps, commands, settings and versions that appear in the excerpts.
General	Answer only from the excerpts.

Table 14: *Domain-specific rules.*

8. Safety and Guardrails

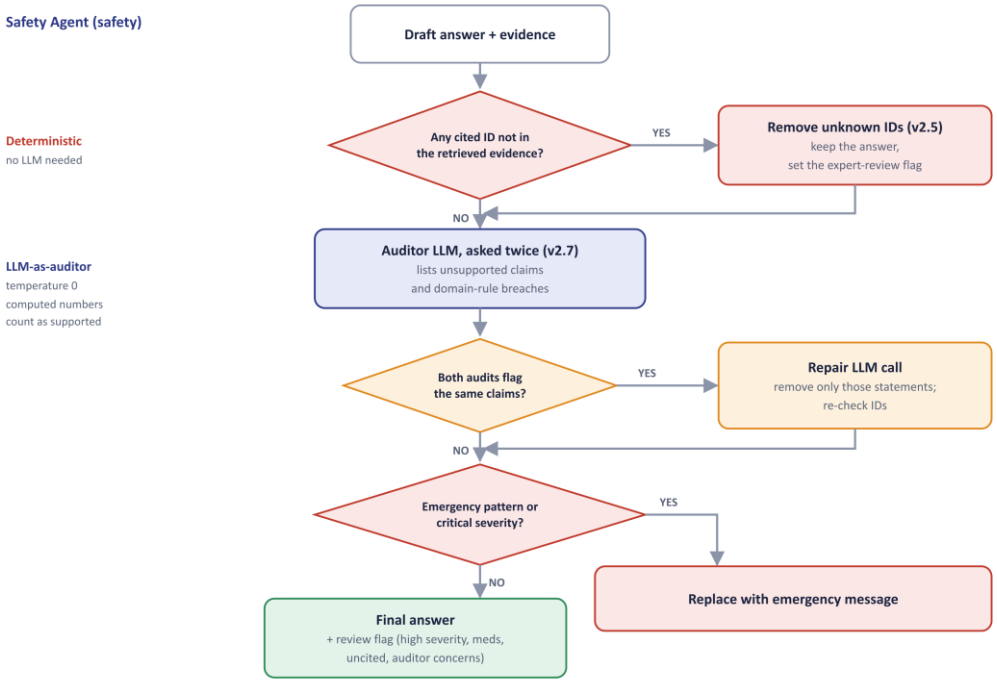


Figure 8: *Safety Agent decision flow (v2.9).*

- **Citation verification (deterministic):** every `[RAG-…]`/`[FDA-…]` ID in the draft must belong to the retrieved evidence. Until v2.4 one unknown ID blocked the whole answer; since **v2.5** unknown IDs are removed and the expert-review flag is set, because a single wrong ID had blocked correct answers (BIO-01).
- **Citation normalisation (v2.1):** gpt-oss writes IDs as 【RAG-0233】, **RAG-0233** or [RAG-1, RAG-2]; these are normalised first, which removed 7 of the 9 false refusals seen in v2.0.
- **Auditor + repair (v2.1, v2.7):** an auditor LLM lists unsupported claims and a repair call removes only those statements (not the whole answer). The auditor is not fully consistent from run to run, so since **v2.7** it is asked twice and a statement is removed only when both audits flag it; numbers computed from cited values (finance calculations) count as supported.
- **Empty replies (v2.9):** if the answer or repair call returns nothing, it is retried with low reasoning effort instead of producing a refusal.
- **Uncited answers** are kept but flagged for expert review rather than blocked.
- **Emergency rule:** six regular expressions (chest pain, can't breathe, anaphylaxis, severe bleeding, limb weakness, loss of consciousness) or a *critical* severity from the Router Agent replace the answer with an emergency-care message.
- **Medication-label check** (Domain Check Agent, health questions only) reports an interaction only when an exact quote from the OpenFDA label supports it, and never claims a combination is safe.
- **Disclaimer:** research prototype only — not medical, legal, financial or technical advice.

9. Version History: v2.0 to v2.9

The system was improved in small steps. After each step the **same 30 questions** were run through the full pipeline and judged by `gpt-oss-120b`, so every version can be compared with the one before; from v2.5 the 18 in-dataset questions were run as well. Only one or two things were changed per version, so the measured effect can be attributed to the change.

9.1 What changed in each version and its effect

Version	What changed	Tested	Hit	Answered	Correct (30)	18-set	Measured effect
v2.0	Baseline: 100 rows/domain, top-5 hybrid search, strict citation check (whole answer blocked on any problem)	34 of 50 (quota)	—	74%	37.5%	—	Routing 81%, answered 74%, correct 37.5% (34 of 50 questions tested before the quota ran out; different question set)
v2.1	Citation normaliser, safety repair (remove only bad statements), domain vote, 200 rows/domain, de-duplication	30 of 30	70%	90%	50%	—	15/30; Routing 30/30; retrieval hit 21/30; legal retrieval 0/6
v2.2	Contract-scoped legal search when the question names a contract	6 legal (retrieval)	90%	—	—	—	Legal retrieval 0/6 → 6/6 (offline check on the 6 legal questions)
v2.3	Colab speed: GPU embedding batches, progress bar, Gradio debug=False	runtime + 9 demo scenarios	—	—	—	—	Runtime only: embedding ≈ 2–3 min on T4; no accuracy change
v2.4	top-k 5 → 8 evidence chunks; retry an empty answer	30 of 30	93%	80%	63%	—	19/30 (+4); Retrieval hit 21 → 28 of 30
v2.5	Finance: add each report's table; error codes get extra keyword matches; Safety Agent removes unknown citations instead of blocking	30 + 18	93%	97%	77%	13 + 2p	23/30 (+4); Finance 1 → 2/6; unknown IDs no longer block answers
v2.6	Finance tables shown with every value labelled by its column	30 + 18	93%	87%	77%	13 + 2p	23/30 (±0); Same total; finance tables read correctly (FIN-04)
v2.7	Safety audit asked twice before removing a statement; computed numbers count as supported	30 + 18	93%	93%	80%	14 + 1p	24/30 (+1); Finance 2 → 3/6; fewer correct answers removed by the audit
v2.8	Named contract: 12 chunks; general questions: second-hop search for the missing fact	30 + 18	93%	93%	83%	11 + 2p	25/30 (+1); General 6/6, support 6/6
v2.9 (final)	Empty model reply retried with low reasoning effort (answer and repair)	30 + 18	93%	97%	83%	12 + 3p	25/30 (±0); Answered 29/30; no answer lost to an empty reply
v3.0 test	Experiment: gpt-oss-120b instead of gpt-oss-20b — same score, 2× cost, so not adopted	30 + 18	93%	100%	83%	12 + 2p	25/30 (±0); Same score at 2× cost and 2–3× time → not adopted

Table 15: *Version history with the measured effect of each change (percentages of the 30-question test; 18-set = correct + partial of the 18 in-dataset questions).*

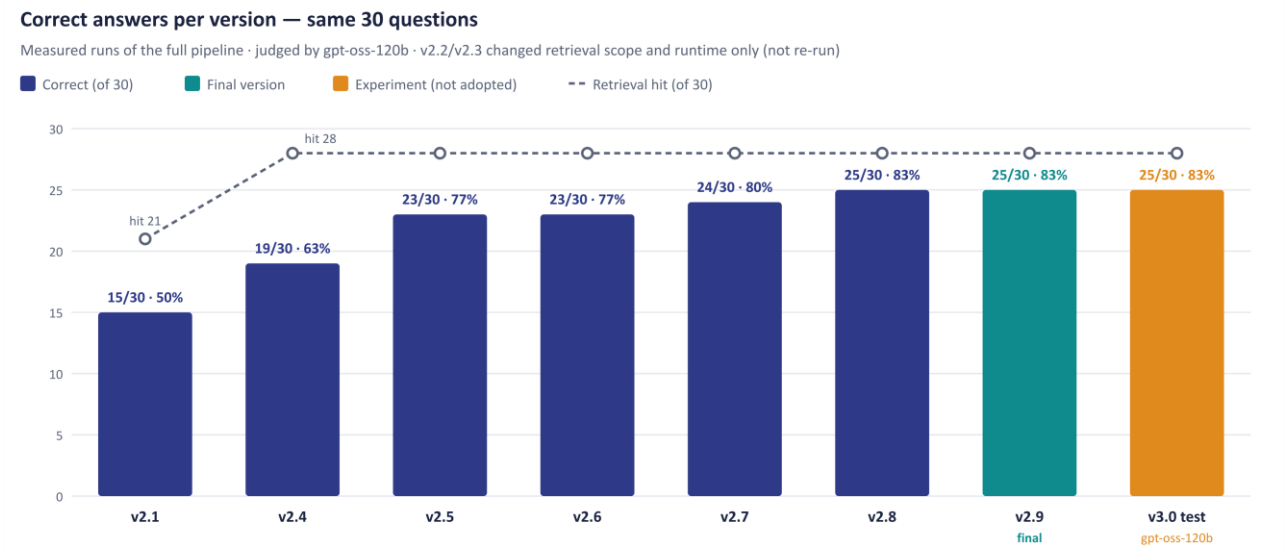


Figure 9: *Correct answers per version on the same 30 questions, with the retrieval hit.*

9.2 Correct answers per domain and version

Domain	v2.1	v2.4	v2.5	v2.6	v2.7	v2.8	v2.9	v3.0 test
Biomedical	3/6	4/6	5/6	4/6	5/6	4/6	4/6	4/6
Legal	2/6	5/6	6/6	6/6	6/6	6/6	6/6	6/6
Finance	1/6	1/6	2/6	2/6	3/6	3/6	3/6	3/6
Customer support	4/6	5/6	5/6	6/6	5/6	6/6	6/6	6/6
General	5/6	4/6	5/6	5/6	5/6	6/6	6/6	6/6
Total (of 30)	15	19	23	23	24	25	25	25
Retrieval hit	21	28	28	28	28	28	28	28
Answered	27	24	29	26	28	28	29	30

Table 16: *Per-domain correct answers for every measured run (v2.2 and v2.3 were not re-run on the full test).*

9.3 Embedding models and retrieval system from the first to the final version

A clear picture of what the retrieval side looked like at each step. The embedding model and chunking stayed fixed; the retrieval logic around them grew from a plain domain-filtered hybrid search to a domain-aware evidence collector.

Version	Embedding & index	Retrieval	Answer & safety	Measured effect
v2.0	MiniLM-L6 (384-d) · 1,200 / 150 chunks · 100 rows/domain · BM25 per domain	Hybrid RRF (k 60), domain filter, top-5	Strict citation check — any problem blocks the answer	Routing 81%, correct 37.5%
v2.1	Same model · 200 rows/domain · de-duplication → 14,704 chunks	+ domain vote (≥ 3 of global top-5) · + global back-stop 2	Citation normaliser · repair instead of block	Routing 100%, hit 21/30, correct 15/30
v2.2	Same	+ contract-scoped search (title match ≥ 0.6)	—	Legal retrieval 0/6 → 6/6
v2.3	Same · GPU batches of 256	—	Gradio debug=False	Index in ≈ 2–3 min on T4
v2.4	Same	top-k 5 → 8 (6 domain + 2 back-stop)	Retry an empty answer	Hit 28/30, correct 19/30

Version	Embedding & index	Retrieval	Answer & safety	Measured effect
v2.5	Same	+ finance report tables (4) · + error-code matches (2)	Unknown IDs removed, not blocking	Correct 23/30, finance 2/6
v2.6	Same	Finance table values labelled by column	—	Correct 23/30
v2.7	Same	—	Two audits must agree · computed numbers supported	Correct 24/30, finance 3/6
v2.8	Same	Named contract 12 chunks · second-hop search (3) for general	—	Correct 25/30
v2.9 (final)	all-MiniLM-L6-v2 · 1,200 / 150 · 14,704 chunks	Hybrid RRF · domain vote · back-stop 2 · top-8 · contract scope · domain add-ons	Empty reply retried with low reasoning effort	Correct 25/30, hit 28/30, answered 29/30
v3.0 test	Same	Same	LLM gpt-oss-120b instead of 20b	Correct 25/30 — not adopted

Table 17: *Embedding model and retrieval system from the start version to the end version.*

Embedding alternatives (`all-MiniLM-L12-v2`, `BAAI/bge-small-en-v1.5`), other chunk sizes, re-ranking and other fusion constants were evaluated in the offline study (Chapters 4–6) and **not** adopted, so they do not appear in the timeline as production changes.

9.4 The v3.0 experiment: a 6× larger model

To test whether the remaining errors are a model limitation, v3.0 replaced the answering model `gpt-oss-20b` with `gpt-oss-120b` and changed nothing else.

Measure	v2.9 · gpt-oss-20b	v3.0 test · gpt-oss-120b
Correct (30 questions)	25/30 (83%)	25/30 (83%)
Answered	29/30	30/30
Retrieval hit	28/30	28/30
18 in-dataset: correct + partial	12 + 3	12 + 2
Cost per question	1× (≈ ₹0.25)	≈ 2×
Answer time	1×	≈ 2–3×

Table 18: *v2.9 versus the v3.0 gpt-oss-120b experiment.*

The larger model answered 30/30 questions, but 4 of its 5 misses are the same questions as in v2.9 (BIO-03, FIN-01, FIN-02, FIN-06): BIO-03 and FIN-02 have faulty reference answers and FIN-01 and FIN-06 name no company — no model can fix that. It got BIO-02 right but did not fully answer BIO-05, which is the same ±1 run-to-run variation seen between versions. The system is limited by the data, not by the model size, so **v2.9 with gpt-oss-20b remains the final version**.

10. Complete Parameter Reference

Every tunable value of every stage — data, chunking, embedding, vector store, keyword search, retrieval, LLM, router, domain check, answer, safety, UI and evaluation — with the final value in v2.9, the values tried or used in earlier versions, the reason the others were not chosen, and the version in which the final value was introduced.

Stage	Parameter	Final value (v2.9)	Values tried / earlier versions	Why not the others	Version
Data	Dataset	RAGBench, 5 subsets	—	Public benchmark with documents + reference answers for 5 domains (CC BY 4.0)	v2.0
Data	Rows per domain (max_rows)	200 (1,000 total)	100 (v2.0), 50 (QUICK_TEST)	100 left gaps in coverage; 200 keeps CUAD at ~10k chunks, embeddable in minutes on GPU	v2.1
Data	De-duplication	On (exact text, indexed once)	Off (v2.0)	Repeated passages filled several evidence slots	v2.1
Chunking	chunk_size	1,200 characters	300, 500, 800, 2,000	Same accuracy band at the lowest cost; 300 needs 5.7× chunks; 2,000 exceeds MiniLM's 256-token window	v2.0
Chunking	chunk_overlap	150 characters (12.5%)	50, 100, 150 (on 300), 200, 300	Keeps boundary sentences in both chunks without doubling the index	v2.0
Chunking	Boundary rule	Last space in 2nd half of window	Hard cut	Never splits a word	v2.0
Embedding	Model	all-MiniLM-L6-v2	all-MiniLM-L12-v2, bge-small-en-v1.5	84% vs best 87% hybrid hit@5; fastest and smallest	v2.0
Embedding	Dimensions	384	384 (all-MiniLM-L12-v2), 384 (bge-small-en-v1.5)	Same size as the alternatives; the speed difference decides	v2.0
Embedding	normalize_embeddings	True	False	Dot product = cosine similarity	v2.0
Embedding	batch_size	256 GPU / 32 CPU	32 everywhere (v2.0–v2.2)	Fills the T4 GPU; ~2–3 min for the full index	v2.3
Vector store	Store	InMemory (Colab) / Chroma (local)	Chroma on Drive	No server; Chroma caches the index between local runs	v2.0
Keyword	BM25 variant	BM25Okapi (k1 = 1.5, b = 0.75)	—	Standard defaults; one index per domain + global	v2.0
Keyword	Tokenizer	lower-case alphanumeric (keeps hyphens)	—	Keeps codes like SQLCODE-1585, CVE-2017-8804	v2.0
Retrieval	Fusion	Reciprocal Rank Fusion	BM25 only, dense only	Complementary failures; best MRR	v2.0
Retrieval	RRF k	60	10, 30, 100	No difference measured; standard value	v2.0
Retrieval	top_k	8	3, 5 (v2.0–v2.3), 10	Study: 5 → 84%, 8 → 87%, 10 → 90%; live hit 21 → 28 of 30 after the move to 8	v2.4
Retrieval	GLOBAL_BACKSTOP	2	0, 1, 3	Guards against Router Agent errors; cost 1 finance question when the domain is known	v2.1
Retrieval	DOMAIN_VOTE_MIN	3 of global top-5	none (v2.0)	Routing 81% → 100% in the live test	v2.1
Retrieval	Contract match	token overlap ≥ 0.6, unique best	substring match	Legal hit@5 20% → 100%	v2.2
Retrieval	LEGAL_TOP_K (named contract)	12	8, 16	Answer sentences found: IN-LEG-03 2/8 → 5/8, LEG-01 0/3 → 3/3; 16 adds almost nothing	v2.8
Retrieval	REPORT_TABLES (finance)	up to 4 table chunks of each retrieved report	none; neighbouring chunks; all sibling documents	FinQA stores a report's table as a separate document; neighbours never reached it. Answer table now given for FIN-03, FIN-04	v2.5

Stage	Parameter	Final value (v2.9)	Values tried / earlier versions	Why not the others	Version
Retrieval	CODE_EXTRA (error codes)	2 extra BM25 matches, appended	fused into RRF	Appending found CUS-02's fix (SQL1585N); fusing pushed out good chunks of CUS-01 and BIO-03	v2.5
Retrieval	FOLLOW_UP_K (general)	3 chunks from a second-hop search	none	Multi-hop: GEN-02 now finds "Gibraltar head of state" → Elizabeth II	v2.8
Retrieval	Re-ranker	None	cross-encoder ms-marco-MiniLM-L-6	+1.93 s/question, no gain over hybrid in-domain search	v2.0
LLM	Model	openai/gpt-oss-20b (Groq)	gpt-oss-120b (v3.0 test), llama-3.3-70b	120b scored the same 25/30 at 2× cost and 2–3× time; llama-3.3-70b not available on the key	v2.0
LLM	temperature	0	—	Reproducible, least creative answers	v2.0
LLM	Empty reply	retry twice with reasoning_effort = low	retry once at default effort (v2.4)	gpt-oss sometimes spends its whole output on reasoning on long technotes	v2.9
LLM	Structured output	json_schema + 3 retries	function calling	gpt-oss mis-cased tool names; schema echo fixed by retry	v2.1
Router	Domain labels	5 + unknown	—	Detailed descriptions per domain in the schema	v2.0
Domain check	MAX_MEDICATIONS / label chars	6 / 3,000	—	Bounds OpenFDA calls and prompt size	v2.0
Answer	Citation format	[RAG-xxxxx] after each fact	free form (v2.0)	Normaliser fixed 7 of 9 false refusals	v2.1
Answer	Finance tables	each value labelled with its column	raw JSON rows (v2.0–v2.5)	Model read "after 2020" instead of "total" (FIN-04); fixed	v2.6
Safety	Unsupported claims	remove only if two audits agree, then repair	block whole answer (v2.0); single audit (v2.1–v2.6)	The auditor is not consistent run to run; one audit blocked correct answers (BIO-06, FIN-05)	v2.7
Safety	Unknown source IDs	removed from the answer, review flag on	block whole answer (v2.0–v2.4)	A wrong ID blocked correct answers (BIO-01)	v2.5
Safety	Emergency patterns	6 regular expressions	—	Deterministic escalation	v2.0
UI	Gradio launch	share=True, debug=False	debug=True	Cell finishes; app keeps running	v2.3
Evaluation	Questions / judge	30 (6 per domain) + 18 in-dataset / gpt-oss-120b	50 (v2.0, quota ran out)	Same 30 questions for every version, so versions are comparable	v2.1

Table 19: *Parameters of all stages.*

11. Evaluation and Results

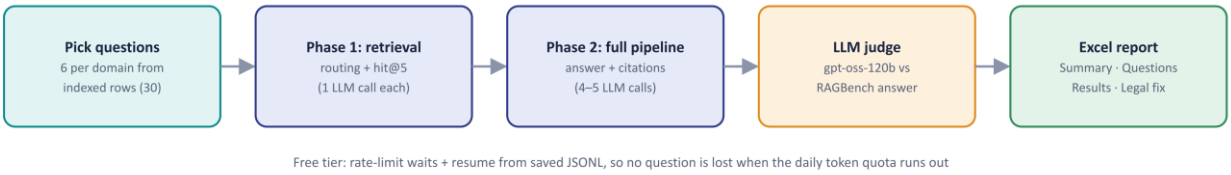


Figure 10: *Evaluation process.*

11.1 Method and metrics

Metric	Formula	Definition	Component	v2.9 result
Routing	routed to the true domain ÷ 30	Router Agent (after the domain vote) chose the question's true domain	Router Agent	30/30 · 100%
Retrieval hit	≥ 1 chunk from the question's own source documents ÷ 30	At least one of the evidence chunks (8 from v2.4) comes from the question's own RAGBench source documents	Retrieval Agent	28/30 · 93%
Answered	answers released, not refused ÷ 30	The Safety Agent released an answer (not a refusal)	Safety Agent	29/30 · 97%
Correct / partial	judge verdict CORRECT ÷ 30	An LLM judge (gpt-oss-120b) rates the answer against the RAGBench reference answer	Answer Agent (end to end)	25/30 · 83%

Table 20: *Evaluation metrics.*

30-question test: six questions per domain drawn from rows that are in the index; the same 30 questions were used for every version. **18 in-dataset questions** (from v2.5): a second set from indexed rows — 4 biomedical, 4 finance, 4 support, 3 legal (with the contract named) and 3 general — graded correct, partial or incorrect. Phase 1 runs the Router and Retrieval Agents; phase 2 runs the full pipeline and the judge; results are saved as JSONL so a run resumes after rate-limit waits.

11.2 Results of the 30-question test (v2.9)

Domain	Routing	Retrieval hit	Answered	Correct
Biomedical	6/6	6/6	5/6	4/6
Legal	6/6	6/6	6/6	6/6
Finance	6/6	4/6	6/6	3/6
Customer support	6/6	6/6	6/6	6/6
General	6/6	6/6	6/6	6/6
Overall	30/30 (100%)	28/30 (93%)	29/30 (97%)	25/30 (83%)

Table 21: *Final v2.9 results by domain (30 questions).*

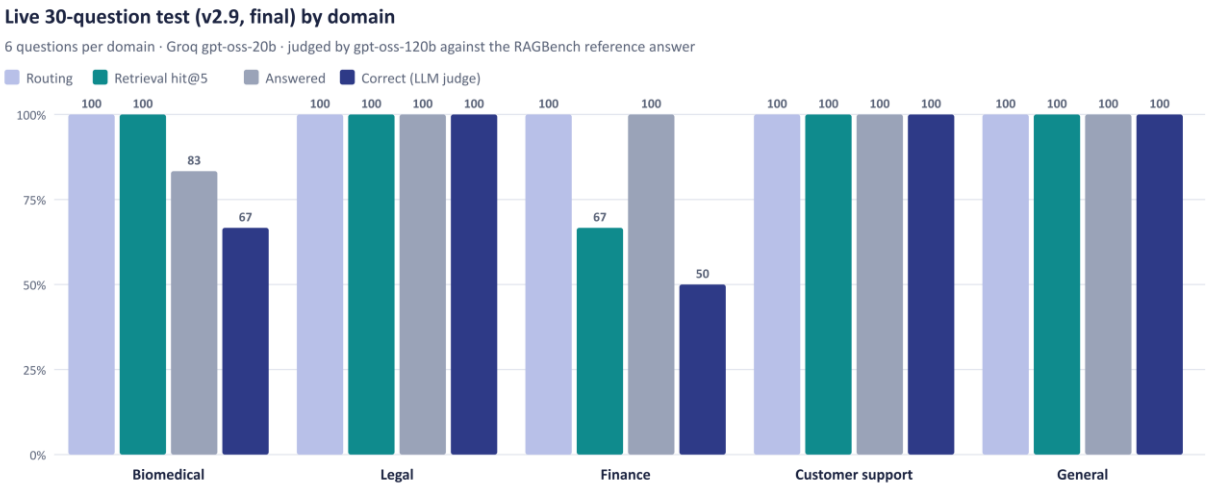


Figure 11: *v2.9 test metrics by domain.*

11.3 Raw, cleaned and answerable scores

Score	Questions counted	Correct	Result
Raw — all 30 questions	30	25	**83%**
Cleaned — 2 faulty reference answers removed (BIO-03, FIN-02)	28	25	**89%**
Answerable — also the 2 questions naming no company removed (FIN-01, FIN-06)	26	25	**96%**
18 in-dataset questions (raw)	18	12 + 3 partial	67%
18 in-dataset, faulty reference removed (IN-GEN-03)	17	12	71%

Table 22: *v2.9 scores — raw and with the faulty or unanswerable questions removed.*

The test questions were **not changed** to raise the score: the cleaned and answerable scores are reported next to the raw score, not instead of it, and every removed question is listed with its reason in Section 11.7.

11.4 Domain-wise results: v2.1 versus v2.9

Domain	Routing v2.1 → v2.9	Hit v2.1 → v2.9	Answered v2.1 → v2.9	Correct v2.1 → v2.9
Biomedical	6 → 6	5 → 6	4 → 5	3 → **4**
Legal	6 → 6	0 → 6	6 → 6	2 → **6**
Finance	6 → 6	4 → 4	5 → 6	1 → **3**
Customer support	6 → 6	6 → 6	6 → 6	4 → **6**
General	6 → 6	6 → 6	6 → 6	5 → **6**
Overall (of 30)	30 → 30	21 → 28	27 → 29	15 → **25**

Table 23: *Per-domain change from v2.1 to v2.9 (questions out of 6 per domain).*

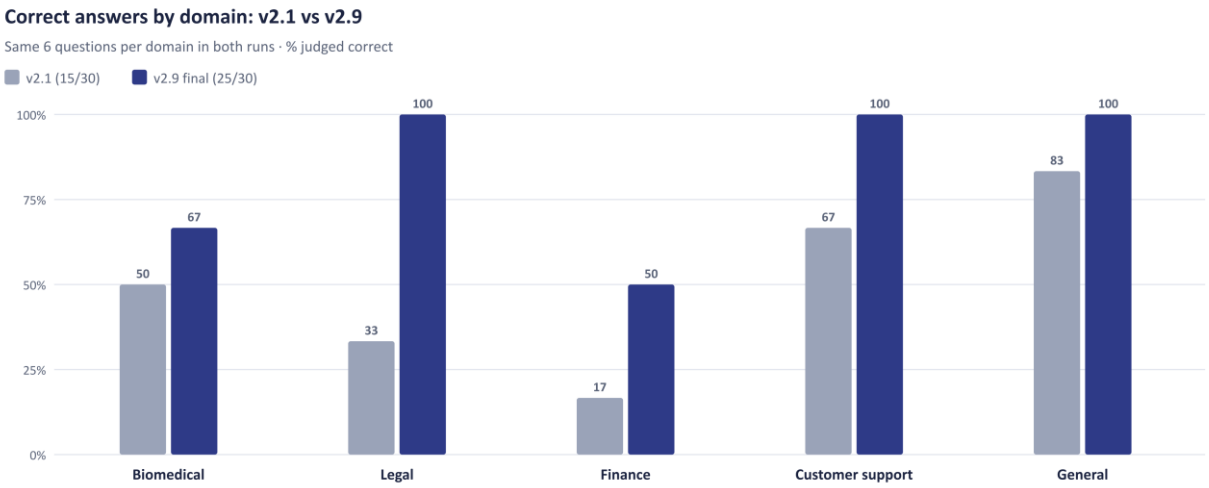


Figure 12: *Correct answers by domain, v2.1 vs v2.9.*

11.5 Legal fix (v2.2)

Check	Before (v2.1)	After (v2.2)
Legal hit@5 (offline, contract scope)	0/6	6/6
Contract name → correct contract (full title)	—	6/6
Company name only → correct contract	—	5/6
Live: Router Agent extracted the name and scoped the search	—	6/6
Live: all chunks from the right contract	—	6/6
Legal correct in the 30-question test (v2.1 → v2.9)	2/6	6/6 (v2.9)

Table 24: *Effect of contract-scoped legal search.*

11.6 The 18 in-dataset questions (v2.9)

Domain	Questions	Routing	Retrieval hit	Answered	Correct	Partial
Biomedical	4	4/4	4/4	4/4	4/4	0
Legal	3	3/3	3/3	3/3	2/3	1
Finance	4	4/4	2/4	4/4	2/4	0
Customer support	4	4/4	4/4	4/4	2/4	2
General	3	3/3	3/3	2/3	2/3	0
Overall	18	18/18	16/18	17/18	12/18 (67%)	3

Table 25: *18 in-dataset questions by domain (v2.9); the question list is in Appendix B.*

11.7 Why the remaining questions fail

Every miss of the final version was checked by hand against the RAGBench source documents and reference answers. Each has one of four causes: a faulty reference answer, a question that names no company, run-to-run variation of the model, or an answer that is only partly complete.

The honest explanation

Of the 5 misses in the 30-question test, **2 have faulty RAGBench reference answers** (BIO-03 asks about MA15 but the reference describes MVA; FIN-02's reference is garbled) and **2 name no company**, so the question fits many of the 200 annual reports in the index. **BIO-02** was correct in every run from v2.4 to v2.8 and missed only in this run — run-to-run variation of the LLM.

In the 18-question set, IN-FIN-03 and IN-FIN-04 name no company, IN-GEN-03 has a reference answer contradicted by its own source, IN-CUS-03 and IN-LEG-03 are correct in their core but partly complete, and IN-CUS-04 was complete in earlier runs (run-to-run variation).

We did **not** change any test question, let the model answer from its own memory, or re-run the test until a lucky result appeared. Results vary by about ±1 question per run because the LLM is not fully deterministic even at temperature 0.

Question	Set	Verdict	Cause	Explanation
BIO-03	30	not answered	Faulty reference answer in RAGBench	The question asks about MA15; the reference answer is about MVA.
FIN-02	30	incorrect	Faulty reference answer in RAGBench	Reference answer is garbled ("$27,867,60").
IN-GEN-03	18	not answered	Faulty reference answer in RAGBench	Reference says neither university has a Philippine campus; the source says the Western Institute of Technology is in Iloilo City, Philippines.
FIN-01	30	incorrect	Question names no company — fits many of the 200 reports	"Change in settlements between 2017 and 2016" — no company named.
FIN-06	30	incorrect	Question names no company — fits many of the 200 reports	"Increase in cash provided by operating activities 2015→2016" — no company named.
IN-FIN-03	18	incorrect	Question names no company — fits many of the 200 reports	"Change in the balance of allowances 2009→2010" — no company named.
IN-FIN-04	18	incorrect	Question names no company — fits many of the 200 reports	"Future minimum lease payments due within 12 months" — no company named.
BIO-02	30	incorrect	Run-to-run variation of the model (correct in earlier runs)	Correct in v2.4–v2.8; this run named the sM2 strain instead of the pgsA-vector strain.
IN-CUS-04	18	partial	Run-to-run variation of the model (correct in earlier runs)	Correct in v2.5, v2.6, v2.7 and the v3.0 test; this run missed part of the reference procedure.
IN-CUS-03	18	partial	Answer partly complete — some of the needed passages not retrieved	Correct core answer (new authorization code needed); misses some steps of the reference.
IN-LEG-03	18	partial	Answer partly complete — some of the needed passages not retrieved	Finds survival, payment and confidentiality clauses; misses IP and non-compete obligations.

Table 26: *Cause of every v2.9 miss.*

Cause	30-question test	18 in-dataset	Can the system fix it?
Faulty reference answer in RAGBench	2	1	No — the reference answer itself is wrong; our answer follows the source document.
Question names no company — fits many of the 200 reports	2	2	Only by asking the user which company is meant (future work); guessing would be dishonest.
Run-to-run variation of the model (correct in earlier runs)	1	1	Partly — LLM output varies about ±1 question per run even at temperature 0.
Answer partly complete — some of the needed passages not retrieved	0	2	Yes, with more context for long procedures and contracts (future work).

Table 27: *Failure causes summarised.*

12. Deployment and Cost

1. Upload `MultiDomain_RAG_Agents_Colab.ipynb` to Colab; choose **Runtime → Change runtime type → T4 GPU**.
2. Add the Groq key under the 🔑 **Secrets** panel as `GROQ_API_KEY` and allow notebook access.
3. **Runtime → Run all.** Install ≈ 1–2 min; download + chunk ≈ 1–2 min; embedding ≈ 2–3 min on GPU.
4. The last cell prints a public `https://….gradio.live` link and finishes (`debug=False`); the app stays live while the runtime is connected.
5. For a quick demo set `QUICK_TEST = True` (50 rows per domain, about 4× faster).

Item	Plan	Effect
Groq gpt-oss-20b	Developer (paid) plan	≈ ₹0.25 per question (4–5 LLM calls); used for the evaluation runs
Groq free tier	8,000 tokens per minute	Enough for a live demo; long test runs wait on rate limits
Groq gpt-oss-120b	Developer plan	≈ 2× the cost per question for the same score (v3.0 test) — not used
Colab T4 GPU	Free, session-limited	Embedding 14.7k chunks in ~2–3 minutes
Gradio share link	Up to 72 hours	Public demo while the Colab runtime runs

Table 28: *Runtime and cost.*

13. Front-End Application

The project includes a web front end (web_app.py with HTML / JavaScript pages) that runs the same five-agent pipeline as the notebook. The screens below were captured from the final version on a desktop browser; the layout also adapts to phone screens.

13.1 Login page

The app opens on a sign-in page. Only users with the demo login can reach the assistant; the status strip shows that the knowledge base is loaded.

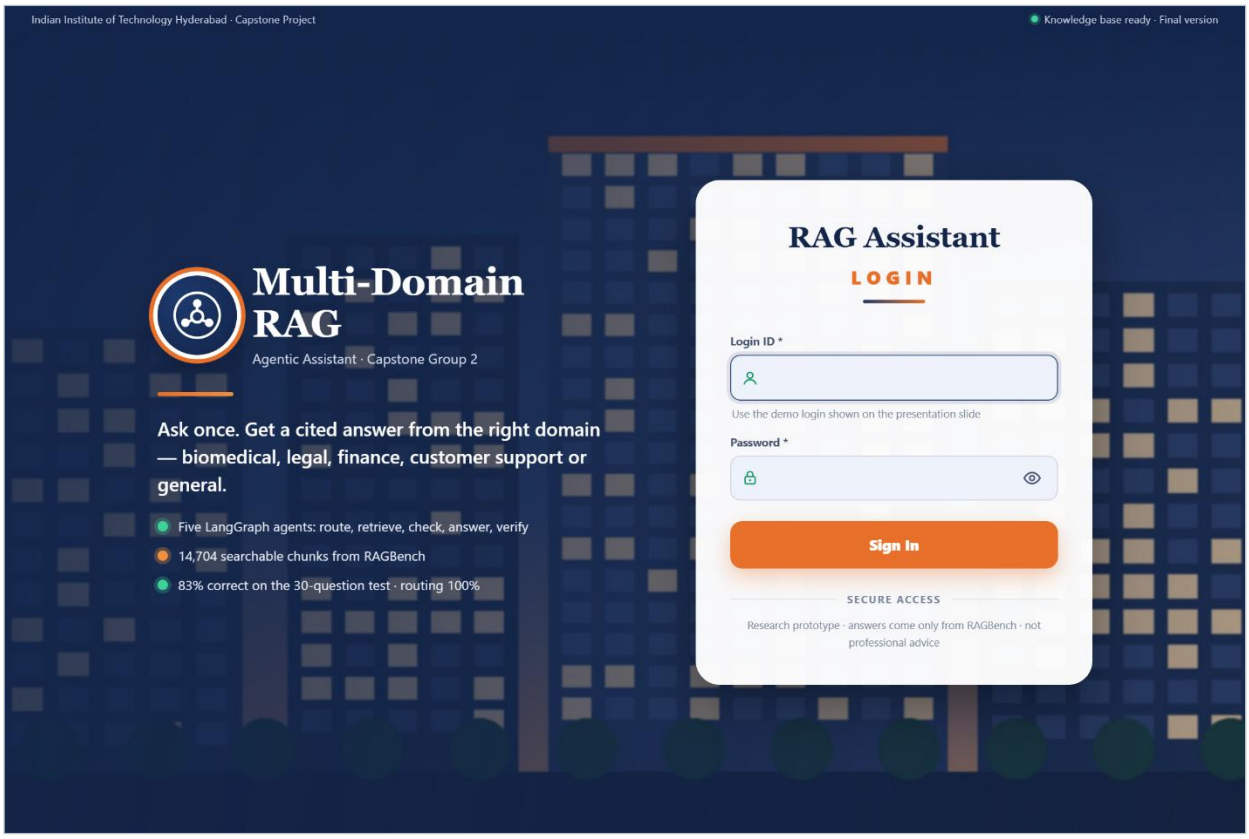


Figure 13: *Login page of the web application.*

13.2 Home screen after sign-in

After signing in, the Assistant tab shows the project summary (5 domains, 1,000 indexed rows, 14,704 chunks, 5 agents, 83% correct), one example question per domain and the search box. The left panel lists recent questions, the five domains and usage tips.

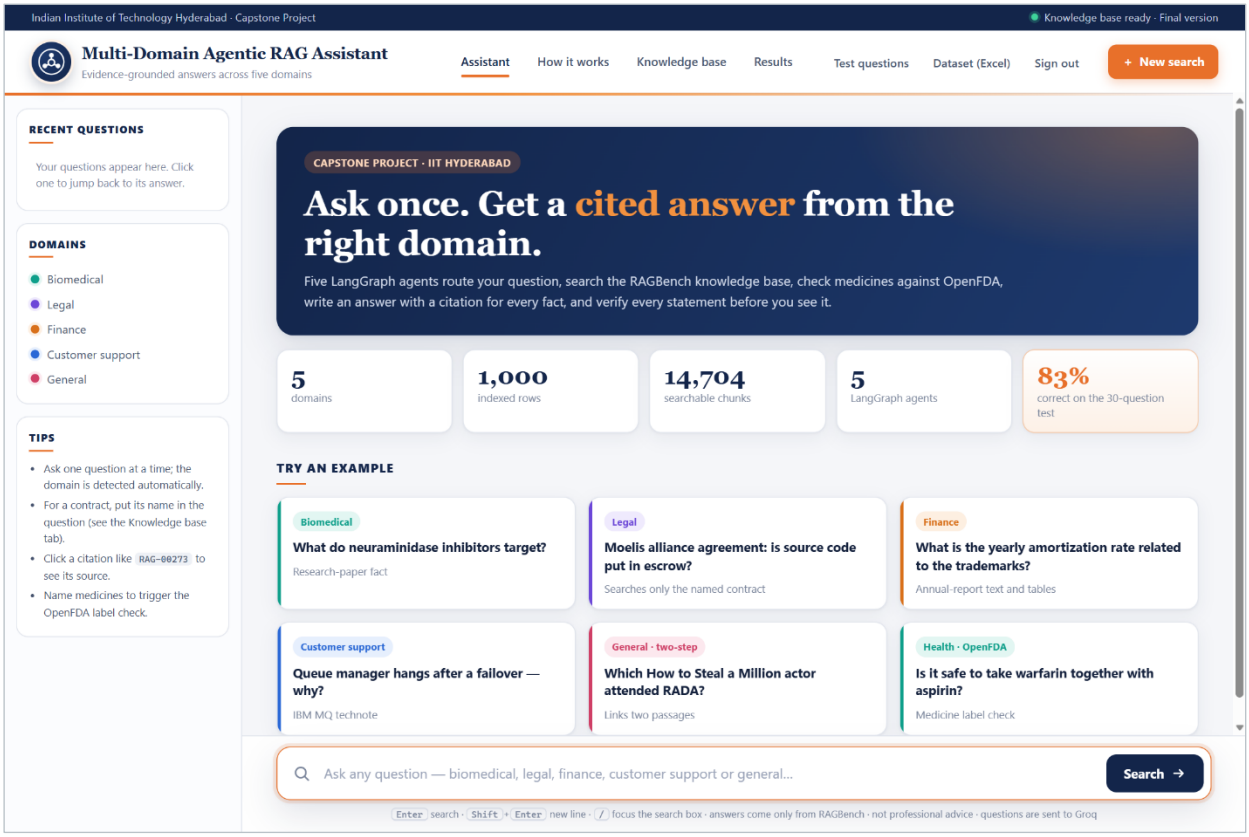


Figure 14: *Home screen after sign-in, with example questions for every domain.*

13.3 Search result: customer support question

A typed question runs through the five agents. The answer card shows the detected domain, severity, number of sources and response time, a progress line for each agent, the answer with a source ID after every fact, and the retrieved sources on the right, with the cited ones marked.

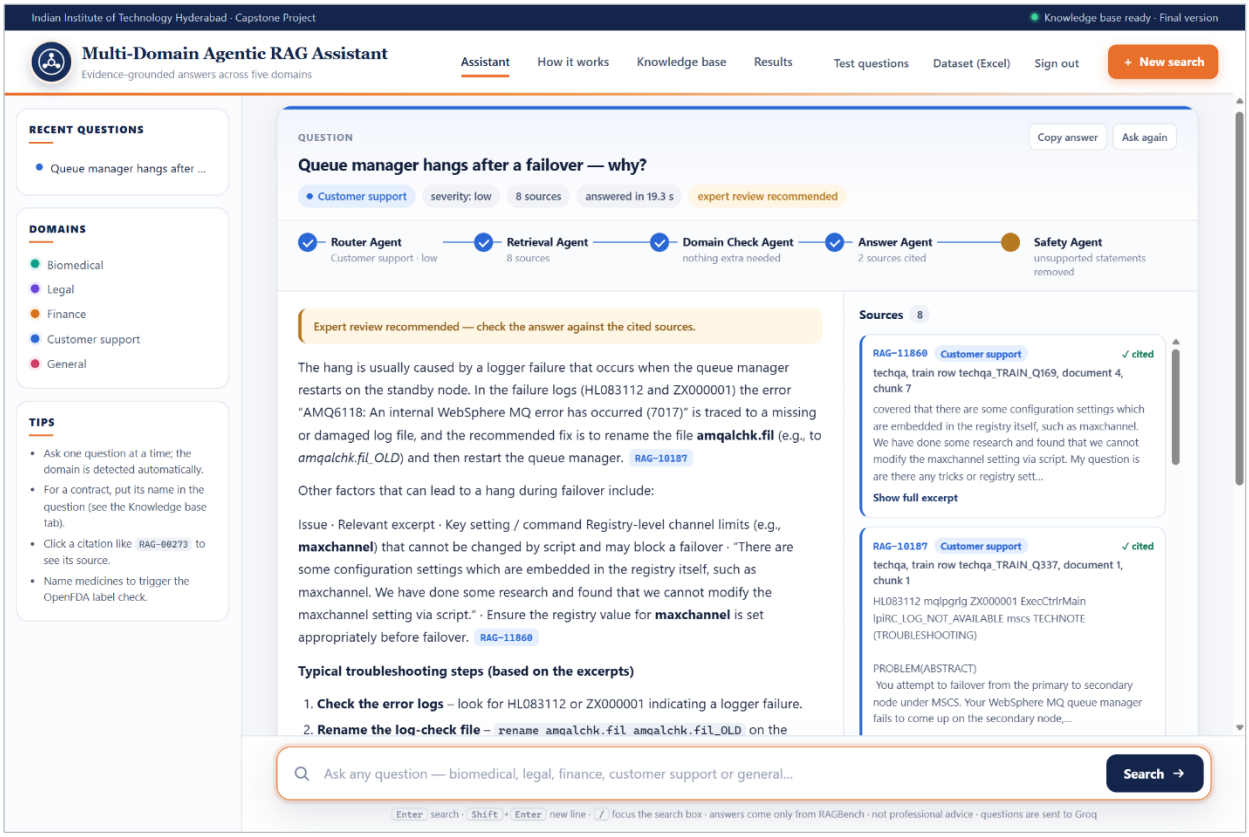


Figure 15: *Answer to a customer-support question, with the agent trace and cited IBM technotes.*

13.4 Search result: medicine question with OpenFDA check

When a question names medicines, the Domain Check Agent fetches their OpenFDA labels (FDA-001, FDA-002). The Safety Agent verifies every statement and flags the answer for expert review, shown as an amber banner.

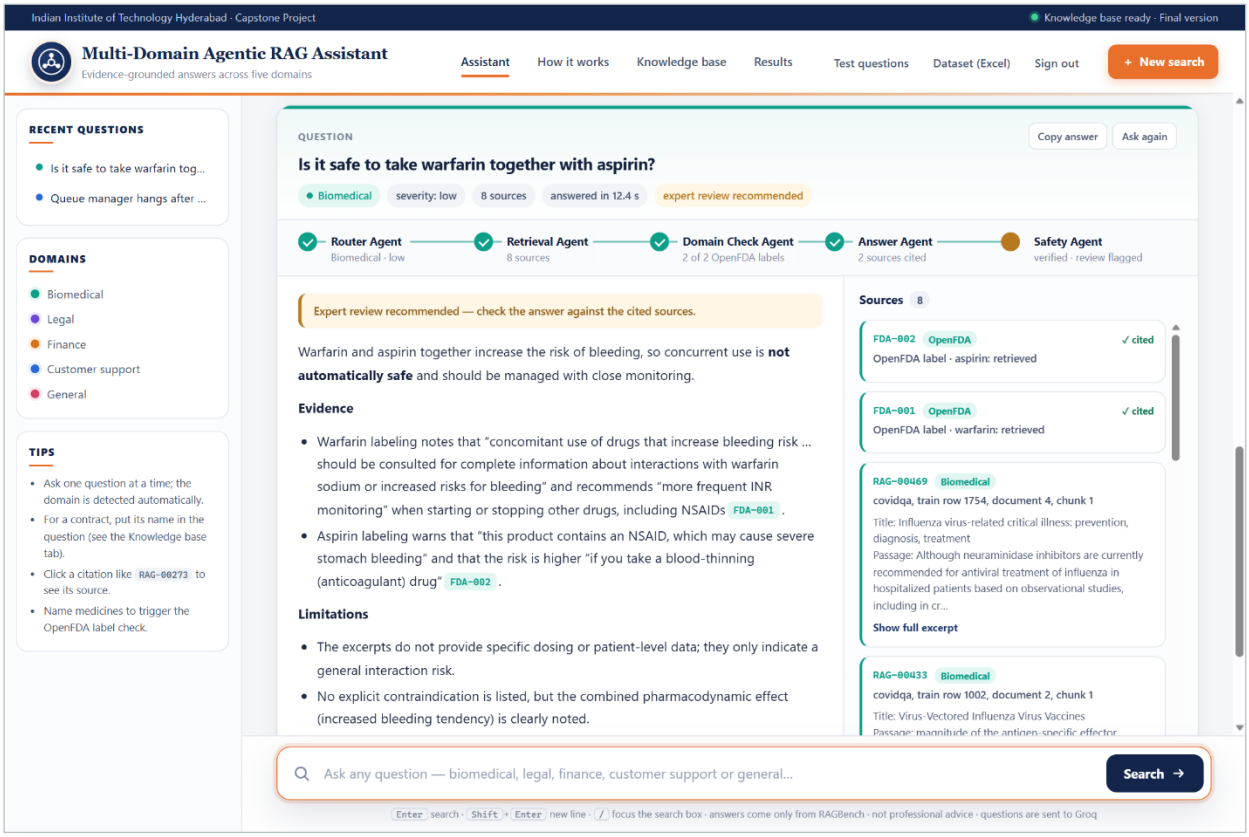


Figure 16: *Medicine interaction question: OpenFDA labels cited and expert review flagged.*

13.5 How it works

This tab explains the five LangGraph agents in order: what each one does and whether it calls the LLM.

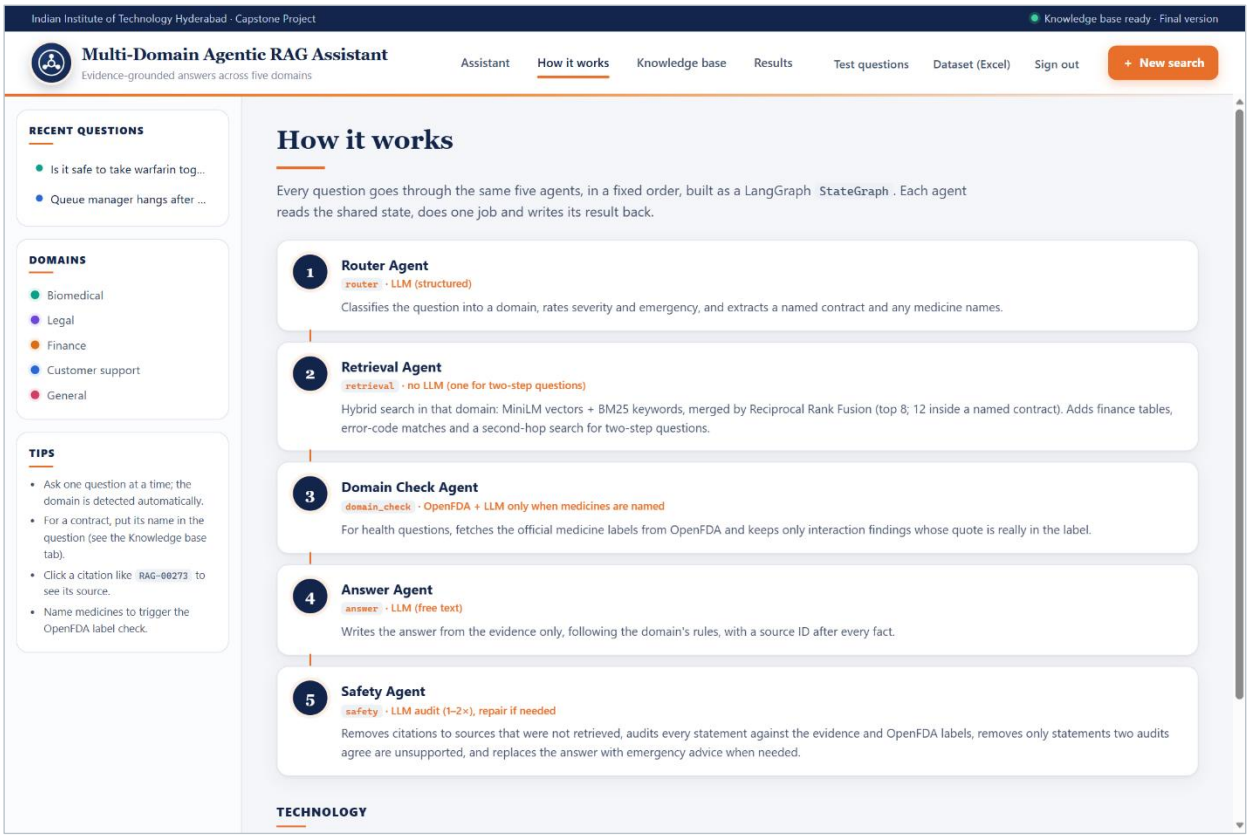


Figure 17: *How it works tab: the five agents of the LangGraph workflow.*

13.6 Knowledge base

This tab describes the five RAGBench sources, the number of chunks and the answer rule for each domain, with a one-click example question.

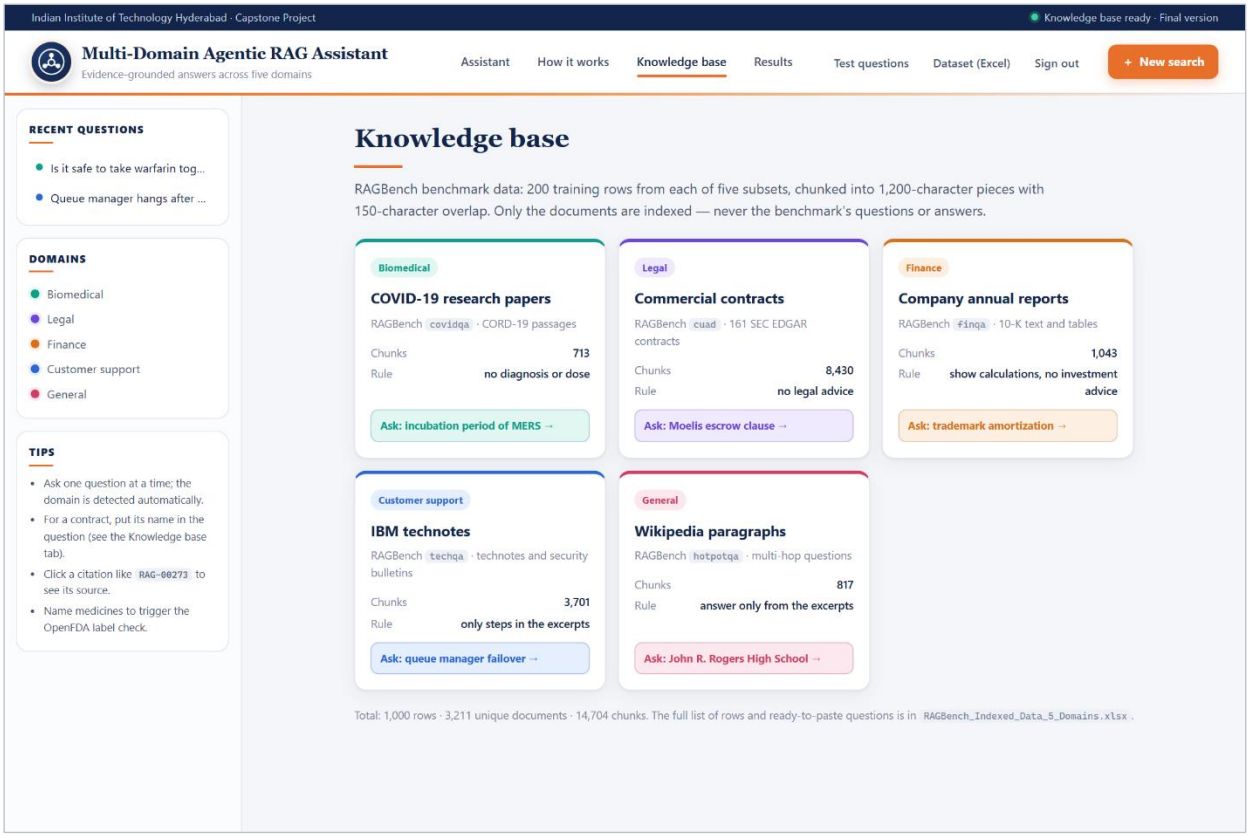


Figure 18: *Knowledge base tab: sources, chunk counts and rules per domain.*

13.7 Results

This tab shows the evaluation results: raw, cleaned and answerable accuracy, routing accuracy, correct answers per version and the final result per domain.

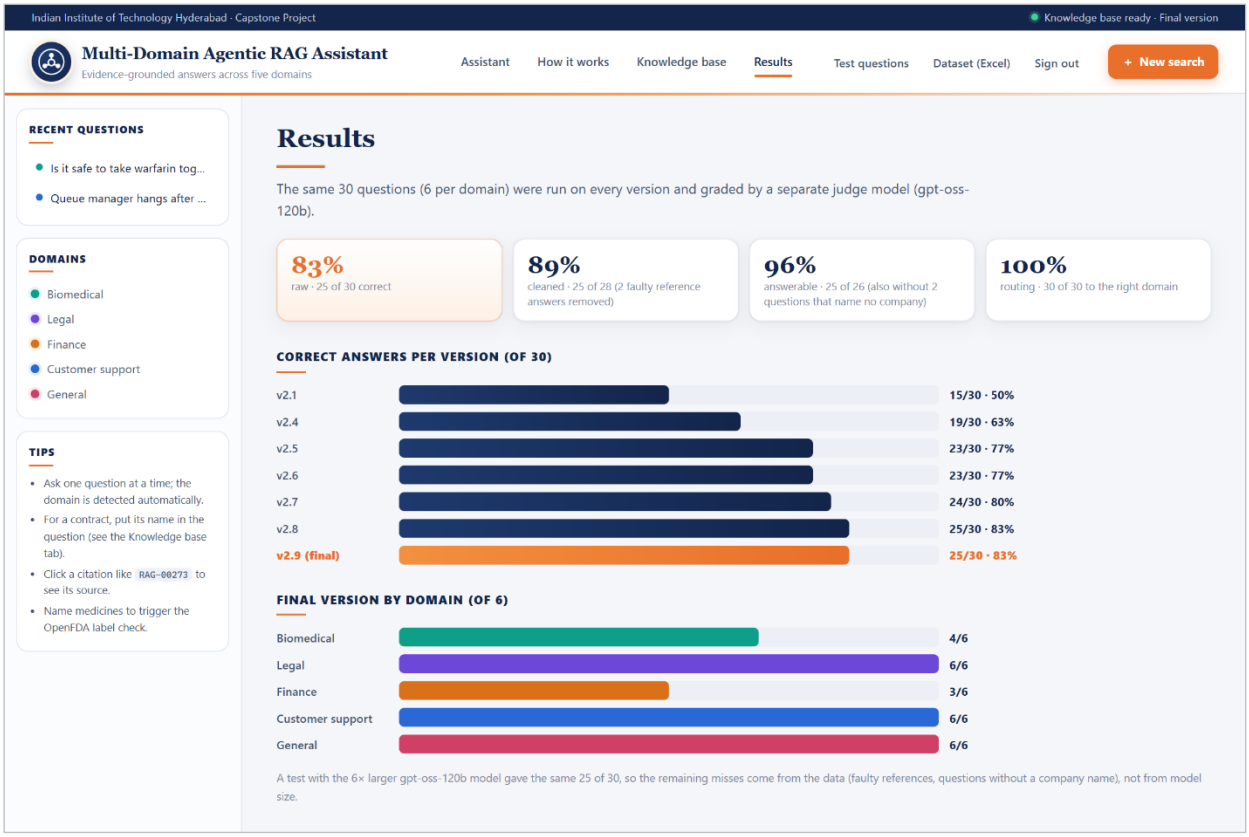


Figure 19: *Results tab: accuracy per version and per domain.*

13.8 Test questions

This page lists the 30 benchmark questions (6 per domain). Ask runs any of them in the assistant, and the indexed data can be downloaded as Excel to try new questions.

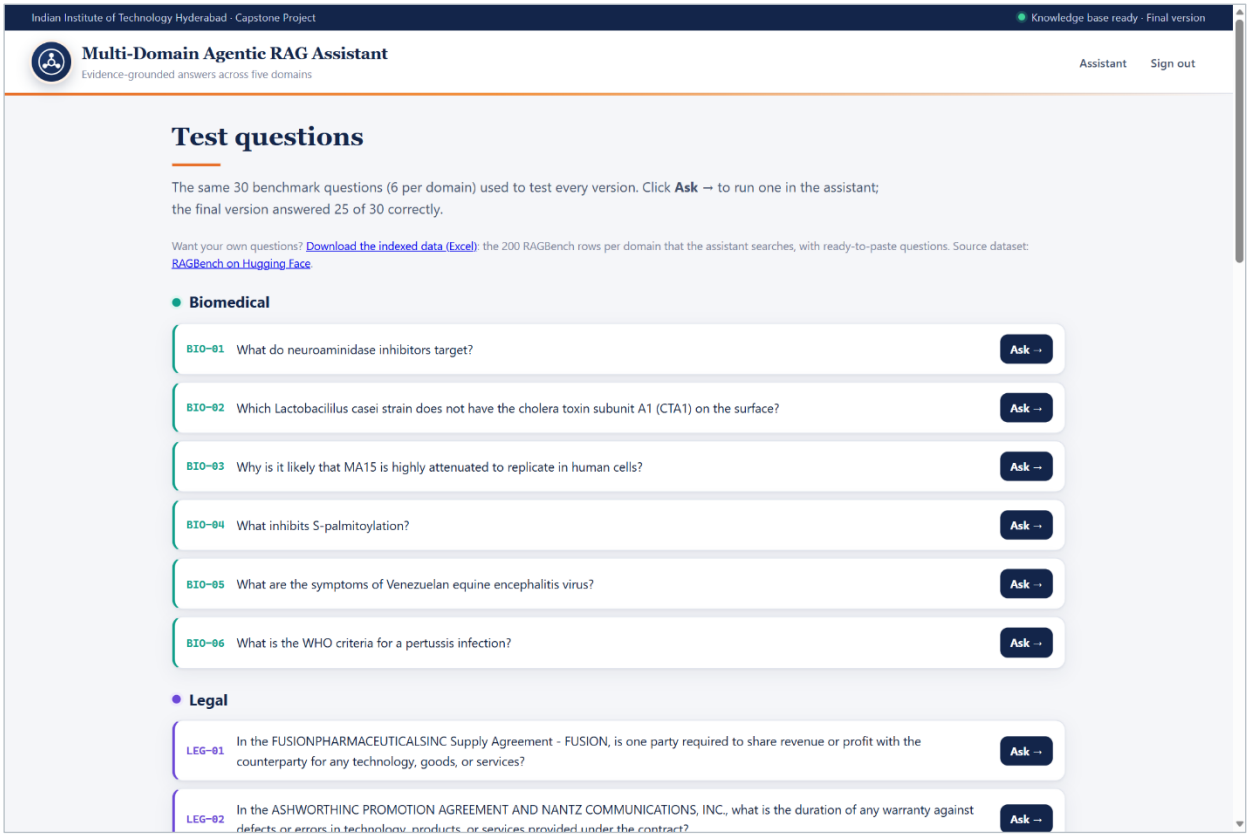


Figure 20: *Test questions page with the 30 benchmark questions.*

14. Limitations and Future Work

- **Questions that name no company (finance):** FIN-01, FIN-06, IN-FIN-03 and IN-FIN-04 fit many of the 200 reports. A company picker (like the contract picker) or a clarifying question would make them answerable.
- **Faulty reference answers:** three RAGBench references (BIO-03, FIN-02, IN-GEN-03) are wrong; a hand-verified test set would measure the system more fairly.
- **Partly complete answers:** long support procedures and contracts (IN-CUS-03, IN-LEG-03) need more of the document; retrieving the whole technote or section, or a re-ranker, could close the gap.
- **Small test set and run-to-run variance:** 30 + 18 questions, one question ≈ 3 points, and ±1 question between runs; averaging several runs over a larger test set would give tighter numbers.
- **Model size is not the bottleneck:** the 6× larger gpt-oss-120b gave the same 25/30 (v3.0 test), so further gains must come from data and retrieval.
- **Embedding upgrade on GPU:** `BAAI/bge-small-en-v1.5` scored best in the parameter study (hybrid hit@5 87% vs 84%) at about 5× the embedding time; with a guaranteed GPU runtime it is the first retrieval upgrade to try.
- **Relevance threshold:** refuse politely when the best retrieval score is low (out-of-knowledge-base questions) instead of answering from weak evidence.
- **Conversation memory:** the assistant currently answers each question independently.

15. Conclusion

The project delivers a working multi-domain agentic RAG assistant that cites every statement and enforces domain-specific safety rules. Measured parameter choices — 1,200-character chunks with 150 overlap, MiniLM-L6 embeddings, hybrid BM25 + dense retrieval with RRF (k = 60), domain-first search with a two-chunk back-stop — gave the best balance of retrieval quality, speed and context size, and the embedding model never needed to change. Ten measured versions then fixed the real failure modes one by one: the domain vote raised routing to 100%, contract-scoped search fixed legal retrieval, top-8 raised the retrieval hit from 21 to 28 of 30, finance report tables and a twice-asked audit raised finance from 1/6 to 3/6, and a second-hop search completed multi-hop answers. The final **v2.9** answers **25/30 (83%)** correctly — 89% on the cleaned test set and 96% on answerable questions — up from 50% in v2.1. Every remaining miss is explained, and a 6× larger model gave the same score, which shows that the remaining limits lie in the data rather than in the model.

References

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2. Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* NeurIPS.
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4. Robertson, S., Zaragoza, H. (2009). *The Probabilistic Relevance Framework: BM25 and Beyond.* Foundations and Trends in IR.
5. Reimers, N., Gurevych, I. (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.* EMNLP.
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7. Hendrycks, D. et al. (2021). *CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review.* NeurIPS Datasets.
8. Chen, Z. et al. (2021). *FinQA: A Dataset of Numerical Reasoning over Financial Data.* EMNLP.
9. Zheng, L. et al. (2023). *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena.* NeurIPS.
10. LangGraph documentation — https://langchain-ai.github.io/langgraph/ ; Groq API documentation — https://console.groq.com/docs.

Appendix A. The 30 Test Questions (v2.9)

ID	Question (shortened)	Routed	Hit	Verdict	Cause if missed
BIO-01	What do neuroaminidase inhibitors target?	✓	✓	correct	
BIO-02	Which Lactobacililus casei strain does not have the cholera toxi …	✓	✓	incorrect	Run-to-run variance
BIO-03	Why is it likely that MA15 is highly attenuated to replicate in …	✓	✓	not answered	Faulty reference
BIO-04	What inhibits S-palmitoylation?	✓	✓	correct	
BIO-05	What are the symptoms of Venezuelan equine encephalitis virus?	✓	✓	correct	
BIO-06	What is the WHO criteria for a pertussis infection?	✓	✓	correct	
LEG-01	In the FUSIONPHARMACEUTICALSINC Supply Agreement - FUSION, is on …	✓	✓	correct	
LEG-02	In the ASHWORTHINC PROMOTION AGREEMENT AND NANTZ COMMUNICATIONS, …	✓	✓	correct	
LEG-03	In the FEDERATEDGOVERNMENTINCOMESECURITIESINC SERVICES AGREEMENT …	✓	✓	correct	
LEG-04	In the Prudential Bancorp Inc Endorsement Agreement, is there a …	✓	✓	correct	
LEG-05	In the BEYONDCOMCORP CO-HOSTING AGREEMENT, is there a clause tha …	✓	✓	correct	
LEG-06	In the MACROGENICSINC COLLABORATION AGREEMENT, is there a fee in …	✓	✓	correct	
FIN-01	what was the change in millions in settlements between 2017 and …	✓	✗	incorrect	No company named
FIN-02	as of december 31 , 2013 , what is the value of securities remai …	✓	✓	incorrect	Faulty reference
FIN-03	what was the change in millions in the computed expected tax fro …	✓	✓	correct	
FIN-04	what was the percent of the total interest on long-term debt to …	✓	✓	correct	
FIN-05	what would total smokeless products shipment volume be in 2017 w …	✓	✓	correct	
FIN-06	what was the percentage increase in the cash provided by operati …	✓	✗	incorrect	No company named
CUS-01	What action is required for IBM WebSphere MQ Hypervisor Edition …	✓	✓	correct	
CUS-02	The configuration task database-transfer failed with DB2 SQL Err …	✓	✓	correct	
CUS-03	OMNIBUS nco_p_mttrapd : Malformed traps How can you determine wh …	✓	✓	correct	
CUS-04	Can an update query be run against the C&DS repository to update …	✓	✓	correct	
CUS-05	Security Bulletin: IBM MQ Appliance is affected by a GNU C libra …	✓	✓	correct	
CUS-06	WAS 8.5.x - Writing a JMS message to a remote queue takes a long …	✓	✓	correct	
GEN-01	Who is a winner of the 2013 6 Hours of Silverstone a co-commenta …	✓	✓	correct	
GEN-02	Who is the head of state of the territory where the Ibrahim-al-I …	✓	✓	correct	
GEN-03	Professional cyclist Sara Symington competed in which Olympic Ga …	✓	✓	correct	
GEN-04	The Minnesota State High School Mathematics League was founded b …	✓	✓	correct	
GEN-05	Which How to Steal a Million actor attended the Royal Academy of …	✓	✓	correct	
GEN-06	Are Chlidanthus and Maianthemum both flowering plants?	✓	✓	correct	

Table 29: *The 30 questions and their v2.9 results.*

Appendix B. The 18 In-Dataset Questions (v2.9)

ID	Question (shortened)	Routed	Hit	Verdict	Cause if missed
IN-BIO-01	Where else MERS-COV has also been detected?	✓	✓	correct	
IN-BIO-02	How long is the SAIBK gene?	✓	✓	correct	
IN-BIO-03	What does immunization with adenovirus induce?	✓	✓	correct	
IN-BIO-04	Where was alpha-tubulin found most abundantly in the cell?	✓	✓	correct	
IN-FIN-01	what was the accumulated pre-tax catastrophe losses from 2010 to …	✓	✓	correct	
IN-FIN-02	what portion of the outstanding shares of our class b common sto …	✓	✓	correct	
IN-FIN-03	what is the percentage change in the balance of allowances from …	✓	✗	incorrect	No company named
IN-FIN-04	what portion of the future minimum lease payments is due within …	✓	✗	incorrect	No company named
IN-CUS-01	What happens if I run arssyscr -I instance_name -r without havin …	✓	✓	correct	
IN-CUS-02	Where can I find information on the Integration Bus Healthcare P …	✓	✓	correct	
IN-CUS-03	How do I transfer my SPSS 24 license key to a new computer? I ne …	✓	✓	partial	Partly complete
IN-CUS-04	How to serve static images from our HTTP Server instead of WebSp …	✓	✓	partial	Run-to-run variance
IN-LEG-01	In the Pareteum Corp — Hosting Agreement, is there a minimum ord …	✓	✓	correct	
IN-LEG-02	In the Zogenix Inc — Distributor Agreement, is there a requireme …	✓	✓	correct	
IN-LEG-03	In the DUOSTECHNOLOGIESGROUP, INC — STRATEGIC ALLIANCE AGREEMENT …	✓	✓	partial	Partly complete
IN-GEN-01	Has Stefan Edberg won more events than Édouard Roger-Vasselin?	✓	✓	correct	
IN-GEN-02	Which team did Lars Lunde play for when defeated for the 1987 Eu …	✓	✓	correct	
IN-GEN-03	Which university offers more campuses in the Philippines, Fairle …	✓	✓	not answered	Faulty reference

Table 30: *The 18 in-dataset questions and their v2.9 results.*

Appendix C. Project Files

File	Purpose
MultiDomain_RAG_Agents_Colab.ipynb	Final notebook (v2.9)
MultiDomain_RAG_Agents_Colab_v2.0 … v2.8_backup.ipynb	Earlier versions, for comparison
Test_Report.docx / .pdf	Test report of the final version and the version history
RAG_30Q_Test_Report.xlsx	30-question test workbook (Summary, Questions Tested, Test Results, Legal Fix v2.2)
MultiDomain_5Domain_Test_Results.xlsx	First 50-question test of v2.0
Retrieval_Parameter_Study.xlsx	Raw results of the chunking / embedding / retrieval experiments
Project_Presentation.pptx	Presentation slides

Table 31: *Project files.*